

UNIVERSIDADE FEDERAL DE SÃO PAULO

DEPARTMENT OF SCIENCE AND TECHNOLOGY

PROFESSIONAL MASTER'S PROGRAM IN TECHNOLOGICAL INNOVATION

**BLOCKCHAIN-ENABLED ABE FRAMEWORK
FOR SECURE IIOT DATA SHARING ACROSS
ORGANIZATIONS**

WESLEY NOVAES FIOREZE COSTA

ADVISOR: PROF. DR. ARLINDO FLAVIO DA CONCEIÇÃO

CO-ADVISOR: PROF. DR. VLADIMIR ROCHA

São José dos Campos – SP

June 2026

UNIVERSIDADE FEDERAL DE SÃO PAULO

DEPARTMENT OF SCIENCE AND TECHNOLOGY

PROFESSIONAL MASTER'S PROGRAM IN TECHNOLOGICAL INNOVATION

**BLOCKCHAIN-ENABLED ABE FRAMEWORK
FOR SECURE IIOT DATA SHARING ACROSS
ORGANIZATIONS**

WESLEY NOVAES FIOREZE COSTA

Dissertation presented to the Professional Master's Program in Technological Innovation at the Universidade Federal de São Paulo, as part of the requirements for obtaining the degree of Master in Sciences.
Advisor: Prof. Dr. Arlindo Flavio da Conceição

São José dos Campos – SP

June 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all the people who supported me directly and indirectly during the stages of this project.

In particular, to my parents, Claudio Souza Costa and Angelita Novaes Fioreze Costa, who sacrificed and motivated me in my academic journey since my childhood. They always instilled confidence, enthusiasm, faith, and love in me. I would also like to honor the memory of my brother who, although no longer with us to witness this, was the first person to encourage me to study Technology in my adolescence. I am also deeply grateful to my partner, Taís Alves Silva Ribeiro, for her motivation and patience throughout this project, as well as for her encouragement to persevere even during turbulent times. The trust you placed in this process was part of what made it possible.

I also express my deep gratitude to the Federal University of São Paulo, an institution I always wanted to be a part of and to which I am proud to belong.

And finally, my sincere thanks to my esteemed advisors, Prof. Dr. Arlindo Flavio da Conceição and Prof. Dr. Vladimir Rocha, for their guidance, patience, and support in the development of this project. Both are great examples for me.

Thank you.

It's like in the great stories, Mr. Frodo. The ones that really mattered. Full of darkness and danger they were. And sometimes you didn't want to know the end. Because how could the end be happy? How could the world go back to the way it was when so much bad had happened? But in the end, it's only a passing thing, this shadow. Even darkness must pass. A new day will come. And when the sun shines it will shine out the clearer. Those were the stories that stayed with you. That meant something, even if you were too small to understand why. But I think, Mr. Frodo, I do understand. I know now. Folk in those stories had lots of chances of turning back, only they didn't. They kept going, because they were holding on to something. That there is some good in this world, and it's worth fighting for.

Samwise Gamgee - The Lord of the Rings

ABSTRACT

The exchange of data among Industrial Internet of Things (IIoT) devices belonging to different organizations is a crucial element for fully exploiting the technological potential of Industry 4.0 and for consolidating this ecosystem. Despite technological advances, significant obstacles related to interoperability and, especially, to the sharing and control of data between entities that do not maintain a prior relationship of trust still persist. In particular, there is difficulty in establishing mechanisms that allow granular control of access to information, while simultaneously ensuring privacy, auditability, and transparency of sharing rules. Traditional solutions tend to depend on centralized intermediaries or implicit trust models, which limits their application in multi-organizational industrial environments. In this context, this dissertation investigates an approach to enable the secure exchange of data in distributed IIoT environments, combining message-oriented communication mechanisms with distributed ledger technologies and cryptographic techniques. The proposal explores the use of smart contracts on the Hyperledger Besu blockchain to formalize, manage, and audit access permissions, as well as the application of Attribute-Based Encryption (ABE) to ensure that shared data is accessible only to entities that meet predefined access policies. The developed architecture reduces the dependence on unilateral control over access rules by recording governance-related information on blockchain, while ABE enforces data confidentiality outside the chain. By combining these mechanisms, the solution supports secure and auditable data sharing among organizations, enabling scenarios such as real-time tracking in supply chains and privacy-preserving industrial data exchange, thereby contributing to the digital transformation of organizations and to the advancement of Industry 4.0.

Keywords: Industrial IoT, Blockchain, Data Privacy, Interoperability, Attribute-Based Encryption.

RESUMO

A troca de dados entre dispositivos da Internet Industrial das Coisas (IIoT) pertencentes a diferentes organizações é um elemento crucial para explorar plenamente o potencial tecnológico da Indústria 4.0 e para a consolidação deste ecossistema. Apesar dos avanços tecnológicos, ainda persistem entraves relevantes relacionados à interoperabilidade e, principalmente, ao compartilhamento e controle de dados entre entidades que não mantêm uma relação prévia de confiança. Em particular, observa-se a dificuldade em estabelecer mecanismos que permitam o controle granular de acesso às informações, assegurando simultaneamente privacidade, auditabilidade e transparência das regras de compartilhamento. Soluções tradicionais tendem a depender de intermediários centralizados ou de modelos de confiança implícitos, o que limita sua aplicação em ambientes industriais multi-organizacionais. Diante desse cenário, esta dissertação investiga uma abordagem para viabilizar o intercâmbio seguro de dados em ambientes IIoT distribuídos, combinando mecanismos de comunicação orientados a mensagens com tecnologias de registro distribuído e técnicas criptográficas. A proposta explora o uso de contratos inteligentes na blockchain Hyperledger Besu para formalizar, gerenciar e tornar auditáveis as permissões de acesso, bem como a aplicação de criptografia baseada em atributos (ABE) para garantir que os dados compartilhados sejam acessíveis apenas a entidades que satisfaçam políticas de acesso previamente definidas. A arquitetura desenvolvida reduz a dependência do controle unilateral sobre as regras de acesso, registrando informações relacionadas à governança na blockchain, enquanto a ABE garante a confidencialidade dos dados fora da cadeia. Ao combinar esses mecanismos, a solução suporta o compartilhamento de dados seguro e auditável entre organizações, possibilitando cenários como rastreamento em tempo real em cadeias de suprimentos e troca de dados industriais com preservação da privacidade, contribuindo assim para a transformação digital das organizações e para o avanço da Indústria 4.0.

Palavras-chave: IIoT industrial, Blockchain; Privacidade de dados; Interoperabilidade; Criptografia Baseada em Atributos.

LIST OF FIGURES

2.1	Attribute-Based Encryption Workflow	21
2.2	Smart contract Flow	27
3.1	Steps of the Literature Review Process	32
4.1	Research strategy	36
4.2	Distribution of Identified Limitations	51
4.3	Evolution of Investigated Publications	52
4.4	Interpretation of Results	52
5.1	Component Diagram	56
5.2	Architecture Design	58
6.1	Encryption time by message	68
6.2	Data size growth by payload size.	71
6.3	ABE-protected payload overhead by payload size.	72
6.4	ABE-protected payload overhead by access policy complexity.	73
6.5	Processing time by access policy complexity.	76

LIST OF TABLES

2.1	Main OpenABE Methods and Their Functions	22
3.1	Association between DSR stages and dissertation activities	31
4.1	Research Questions	35
4.2	Inclusion Criteria	37
4.3	Exclusion Criteria	38
4.4	Systematic Literature Review Results: Blockchain and ABE Applications in IoT and Data Privacy	46
5.1	Data representations considered in the overhead evaluation.	62
6.1	Percentage variation in encryption time relative to the baseline configuration (1 attribute).	69
6.2	Accumulated overhead for real sensor measurements.	75

LIST OF ACRONYMS

AA – *Attribute Authority*

ABE – *Attribute-Based Encryption*

ABI – *Application Binary Interface*

ACL – *Access Control List*

AES – *Advanced Encryption Standard*

AI – *Artificial Intelligence*

API – *Application Programming Interface*

CP-ABE – *Ciphertext-Policy Attribute-Based Encryption*

CPU – *Central Processing Unit*

CasDL – *Cascade Deep Learning*

DApp – *Decentralized Application*

DID – *Decentralized Identifier*

ECIES – *Elliptic Curve Integrated Encryption Scheme*

EIF – *European Interoperability Framework*

EVM – *Ethereum Virtual Machine*

IIoT – *Industrial Internet of Things*

IPFS – *InterPlanetary File System*

ISO – *International Organization for Standardization*

IoT – *Internet of Things*

JSON – *JavaScript Object Notation*

KMS – *Key Management System*

KP-ABE – *Key-Policy Attribute-Based Encryption*

MQTT – *Message Queuing Telemetry Transport*

NVMe – *Non-Volatile Memory Express*

OS – *Operating System*

RAM – *Random Access Memory*

RBAC – *Role-Based Access Control*

RPC – *Remote Procedure Call*

SSD – *Solid-State Drive*

SSI – *Self-Sovereign Identity*

TLS – *Transport Layer Security*

USK – *User Secret Key*

VC – *Verifiable Credential*

zk-SNARK – *Zero-Knowledge Succinct Non-Interactive Argument of Knowledge*

CONTENTS

LIST OF ACRONYMS

CHAPTER 1 – INTRODUCTION	13
1.1 Motivation	14
1.2 Problem Statement	16
1.3 Objectives	17
1.3.1 General objective	17
1.3.2 Specific objectives	17
CHAPTER 2 – THEORETICAL FOUNDATION	18
2.1 Cross-Organizational Interoperability for Secure Data Sharing	18
2.2 Attribute-Based Encryption (ABE)	19
2.2.1 CP-ABE $\times$ KP-ABE	22
2.2.2 CP-ABE Use Cases in Distributed Scenarios Across Organizations . . .	23
2.3 Blockchain	24
2.3.1 Blockchain Pillars	25
2.3.2 Smart Contracts	26
2.3.3 Hyperledger Besu	27
CHAPTER 3 – METHODOLOGY	29
3.1 Design Science Research	29

3.2	Systematic Literature Review	32
3.3	Experimental Plan	33
CHAPTER 4 – SYSTEMATIC LITERATURE REVIEW		34
4.1	Research Questions	34
4.2	Literature Search Strategies	36
4.3	Inclusion Criteria	37
4.4	Exclusion Criteria	38
4.5	Results	39
4.5.1	Specific approaches to privacy and security	39
4.5.2	Interoperability Architectures for Secure Data Sharing	43
4.6	Discussion	45
4.6.1	Study Limitations	45
4.7	Conclusion	53
CHAPTER 5 – SYSTEM ARCHITECTURE		54
5.1	Use Case	55
5.2	Proposed Architecture Design	55
5.2.1	Sequence Diagrams	57
5.3	Key Technologies	61
5.4	Experimental Environment	61
CHAPTER 6 – RESULTS		64
6.1	System Integration and Proof-of-Concept Validation	64
6.2	Cryptographic Performance	67
6.3	Cryptographic Overhead Evaluation	69
6.3.1	Spatial Overhead by Payload Size	70
6.3.2	Impact of Access Policy Complexity	72

6.3.3	Accumulated Overhead with Real Sensor Measurements	74
6.3.4	Processing Time Analysis	76
6.4	Privacy and Access Control	77
6.5	Scalability and Data Volume	77
6.5.1	Cryptographic Scalability	78
6.5.2	Data Volume Scalability	78
CHAPTER 7 – DISCUSSION		79
7.1	Interpretation of Results	79
7.2	Advantages of the Proposed Solution	80
7.3	Limitations and Challenges	81
7.3.1	Attribute Revocation	82
7.3.2	Scope of Cryptographic Scalability Assessment	83
CHAPTER 8 – CONCLUSIONS		85
8.1	Conclusions	85
8.2	Contributions to Industry and Academia	85
8.3	Future Work	86
8.4	Research Artifact	87
REFERENCES		89

Chapter 1

INTRODUCTION

The growth of Industry 4.0 has driven a structural change in manufacturing, characterized by the integration of cyber-physical systems, automation, and digital technologies in industrial environments. The authors Kagermann et al. (2013) describe Industry 4.0 as the deep integration of physical production systems with digital infrastructures, enabling interconnected and data-driven industrial processes.

The conceptual origin of the term *Industrial Internet* is commonly associated with Evans & Annunziata (2012), who described the convergence of industrial machines, advanced computing, data analytics, low-cost sensing, and Internet connectivity as a new wave of industrial innovation. In this context, the expression *Industrial Internet of Things* (IIoT) emerged as a specific designation for the application of IoT principles within industrial environments. To establish a formal baseline, this work adopts the definition provided by Industrial Internet Consortium (2017), which characterizes IIoT as a set of interconnected industrial systems and devices capable of collecting and exchanging data to support monitoring and automation.

Since IIoT environments are based on the interconnection of industrial systems and the exchange of operational data, interoperability becomes a central concern. However, interoperability is a multidimensional concept. The European Interoperability Framework (EIF) organizes interoperability into technical, semantic, organizational, and legal layers (European Commission, 2017). ISO 11354-1, published by the International Organization for Standardization (ISO), in turn, structures enterprise interoperability according to aspects such as business, processes, services, and data, as well as conceptual, technological, and organizational barriers (ISO, 2011). Thus, interoperability does not refer only to the exchange of messages between systems, but also to the ability of different entities to coordinate processes, interpret data, and govern interactions across organizational boundaries.

In this dissertation, interoperability is addressed within a restricted scope. The work focuses on technical interoperability, supported by publish/subscribe communication based on

MQTT, and, primarily, on governance- and security-oriented interoperability, aimed at enabling data exchange between independent administrative domains without presupposing a prior trust relationship. Semantic interoperability, including the harmonization of sensor data models and the interpretation of heterogeneous measurements and distinct physical phenomena, is outside the scope of this research.

Recent forecasts indicate that the global number of IoT-connected devices will reach 29 billion by 2030 (ABUSERRIEH; ALALFI, 2024). However, the rapid expansion of industrial systems also intensifies challenges related to secure cross-organizational data sharing, including technical connectivity, access governance, privacy, and auditability (HAZRA et al., 2021). In these scenarios, there is generally a need for real-time data exchange to support decision-making, while each organization maintains its own governance structures, information management flows, and confidentiality requirements. These factors can restrict inter-organizational collaboration, especially when sensitive operational information must be shared without compromising confidentiality and integrity.

The strategic sharing of information may enable new business models, but it also contrasts with the risk of undue data exposure. Therefore, these conditions require mechanisms capable of enforcing secure and manageable data sharing between different organizations, while preserving the ability of each party to define, enforce, and audit its own access rules.

To illustrate this situation, consider a supply-chain scenario involving a sensor manufacturer, an automobile manufacturer, and a maintenance service provider. The maintenance provider may require access to operational data to perform predictive maintenance services. At the same time, the automobile manufacturer may need to restrict production metrics, while the sensor supplier may protect proprietary device configurations. This scenario shows that organizations may need to share data selectively while remaining administratively independent and without relying on a single trusted intermediary.

This need for data sharing between organizations can be supported by mechanisms that ensure granular access control over shared data, without requiring all parties to depend on a single centralized authority.

1.1 Motivation

As highlighted by Mahendran et al. (2025), the lack of mechanisms for secure cross-organizational IIoT data sharing, coupled with the need for access governance and auditability, remains a significant challenge in Industry 4.0, especially among segregated organizations.

Nishi et al. (2024) also emphasize that IoT sensor networks based on standardized architectures such as IEEE P1451.1.6, a smart transducer network standard that facilitates interoperability in distributed systems, still face persistent problems related to secure authentication, trust management, and integration across distinct environments.

Where devices operate under different administrative domains and governance policies, these challenges intensify. Beyond operational limitations, this gap compromises data consistency and reliability, important requirements for feasibility analysis of different data-driven approaches and potential Artificial Intelligence (AI) applications. Traditional IIoT security architectures rely on centralized communication models based on MQTT brokers, static Access Control Lists (ACLs), Transport Layer Security (TLS) authentication, and public key infrastructure-based mechanisms. However, these approaches have limitations in inter-organizational environments where different entities need to share data while preserving granular and dynamic privacy policies. Hamad et al. (2023) explain that MQTT publish/subscribe architectures are inherently dependent on centralized brokers and that their respective TLS-based protection methods primarily protect communication channels but do not meet end-to-end, data-centric protection requirements for the payload itself. As a result, intermediate components, such as brokers, can still access or manage sensitive information during message routing.

Furthermore, Xu et al. (2018) argue that traditional access control approaches, including ACL and RBAC-based models, face scalability and management limitations in ecosystems composed of multiple administrative domains. As highlighted by the authors, the obstacles related to maintaining authorization rules arise as the number of devices, organizations, and access relationships increases. This limitation is particularly relevant in interoperable IIoT scenarios, where different companies may operate under independent governance policies.

To address these challenges, Rasori et al. (2022) describe Attribute-Based Encryption as a data-centric security mechanism capable of incorporating granular access control policies directly into encrypted data. The authors emphasize that this approach reduces reliance on centralized trust models and allows for the secure sharing of information even in distributed or partially untrusted environments, making ABE suitable for applications requiring granular privacy preservation.

In addition to interoperability challenges, data security and privacy remain central concerns in IIoT environments. The distributed and interconnected nature of these environments increases their exposure to cyber threats. According to Check Point Research, cyberattacks targeting industrial IoT devices increased by 45% in 2024 compared with the previous year, exposing industries to risks such as data breaches, unauthorized access, and operational disruptions (Check

Point Research, 2024). Conventional security approaches, often based on centralized control mechanisms, are insufficient to simultaneously guarantee data privacy, scalability, and resilience against single points of failure. In typical manufacturing environments, with processes involving inventory and demand management, interoperability between different companies introduces additional challenges related to traceability, integrity, and architectural performance. Organizations that frequently need to share sensitive operational data with each other require access management mechanisms, information privacy tools, and assurance that the contractual rules of the business will be maintained.

Blockchain technology, adopted in an integrated way with Attribute-Based Encryption methods, has been highlighted in studies as a promising approach to challenges such as these (OUADDAH et al., 2021). Given the fundamental characteristics and pillars of this technology, which will be explored in this work, and also using the behavior of the ABE method, an architecture was proposed so that the parties involved can manage access to the data in an interoperable and granular way.

1.2 Problem Statement

Modern Industrial Internet of Things (IIoT) architectures predominantly rely on centralized entities to ensure privacy and manage data sharing among different organizations. While effective in controlled environments, this model presents limitations in multi-organizational scenarios, including single points of failure, limited auditability, and a lack of trust transparency among independent stakeholders.

At the same time, traditional access control mechanisms—such as Role-Based Access Control (RBAC), Access Control Lists (ACLs), and identity-centric approaches—have often proven insufficient to support the dynamic and heterogeneous nature of IIoT ecosystems, as they were originally designed for closed and static environments and lack the scalability and adaptability needed to handle open, distributed, and multi-party IoT scenarios, as demonstrated in recent studies conducted by Qiu et al. (2020).

As a result, and consequently the central problem addressed in this work, there is currently no consolidated and widely adopted mechanism capable of providing decentralized, granular, and auditable access control for secure cross-organizational data sharing in IIoT environments. This limitation frequently forces companies to adopt excessively permissive access policies or rigid configurations, both with a negative impact on security, privacy, and controlled data sharing across organizational boundaries. This research seeks to fill this gap by investigating archi-

tructures and proposing a mechanism that enables decentralized, attribute-based, and auditable control over data sharing in IIoT.

1.3 Objectives

1.3.1 General objective

To propose an architectural solution for secure cross-organizational IIoT data sharing, focused on data privacy, so that different companies can strategically share data with each other, in different business models, while maintaining independent governance of their information assets.

1.3.2 Specific objectives

- To investigate the current literature, using a systematic review protocol, to identify recently adopted approaches, advances proposed by academia, find limitations in experimental results related to data sharing and privacy, and contribute to advancements in the state of the art of these topics.
- Using a publish/subscribe architecture, implement and test a cryptographic module that will operate in IIoT ecosystems, ensuring data privacy in a refined and adaptable way.
- Design a decentralized layer that will ensure the auditability of granted access permissions and manage the distribution of access to cryptographic keys.
- Experimentally evaluate the performance of the proposed solution under different message sizes and attribute policies.

To achieve the specific objectives outlined, the next chapter will focus on justifying the technological and methodological choices adopted. Initially, the topic of interoperability will be discussed to contextualize the challenges of integrating different organizations. Subsequently, a presentation on the chosen cryptographic method will be presented. Finally, the concept of blockchain, as well as its pillars, will be reviewed, focusing on its role in the project's infrastructure.

Chapter 2

THEORETICAL FOUNDATION

The technologies underpinning this research encompass some of the pillars of modern industry: the Industrial Internet of Things (IIoT), advanced cryptographic methods such as Attribute-Based Encryption (ABE), blockchain technology, and message-oriented protocols such as MQTT.

The MQTT protocol enables message coordination between industrial devices, while ABE provides a cryptographic mechanism capable of enforcing access policies. Blockchain, in turn, introduces decentralization, immutability, and auditability, ensuring that business rules and shared data cannot be changed without consensus. Together, these technologies establish a technology stack upon which the proposed cross-organizational data-sharing architecture is built. The following sections detail each of these components and their relevance to the proposed solution.

2.1 Cross-Organizational Interoperability for Secure Data Sharing

To overcome barriers related to the integration of IIoT devices between different organizations, in their studies, Sychowiec & Zielinski (2023) proposed a platform that integrated Blockchain with Apache Kafka, a distributed data streaming technology. The use of specific broker topics and partitions, integrated with distributed solutions, as detailed by Helu et al. (2020), demonstrates potential scalability and performance improvement, allowing for more efficient communication between devices. However, despite the observed improvements and the analysis of the results published by the authors, obstacles related to technology integration, large-scale implementation, and latency testing were highlighted.

As discussed by Afrin et al. (2025), in cases where companies have divergent policies on data governance rules, integration between such organizations is often hindered. The authors emphasize that IIoT implementations involving very different corporations result in fragmen-

ted architectures, limiting coordinated decision-making. These findings indicate that interoperability in multi-organizational environments can go beyond technical standardization, also involving well-defined architectural strategies and data access policies within these ecosystems.

The PRISM-IIoT framework, proposed by Mahendran et al. (2025), comprises a multi-layered architecture that integrates cryptographic techniques with blockchain. The framework incorporates cryptography-based data protection, access control driven by internal policy groups, and a distributed ledger layer responsible for the immutability of transactions and events. Mahendran et al. (2025) highlighted in their study that, in addition to structural technological challenges of data sharing in 4.0 environments, distributed domains require high levels of privacy. As the authors proposed, by combining a privacy-preserving cryptographic application with tamper-resistant logging mechanisms, PRISM-IIoT mitigates risks associated with unauthorized access, policy violations, and governance inconsistencies across domains. The authors argue that interoperability and multi-stakeholder ecosystems essentially depend on the integration of such decentralized trust mechanisms.

2.2 Attribute-Based Encryption (ABE)

Encryption techniques based on Attribute-Based Encryption (ABE) have proven to be a relevant and promising approach for granular access control in distributed systems, especially in contexts involving multiple agents. Unlike traditional public-key cryptography schemes, ABE allows access to encrypted data to be controlled based on attributes associated with users, supporting fine-grained access control policies.

However, in the context of the Industrial Internet of Things (IIoT), implementing ABE presents practical challenges due to the computational limitations of edge devices, the complexity of attribute revocation, and the additional processing, communication, and storage overhead introduced by cryptographic protection.

The computational and energy cost of ABE is also relevant in IoT and IIoT environments. Wang et al. (2014) evaluated KP-ABE and CP-ABE considering execution time, data and network overhead, energy consumption, CPU usage, and memory usage on heterogeneous devices. Their results show that ABE performance is affected by the number of attributes and the adopted security level. In the evaluated Android smartphone, execution times were more than ten times higher than those observed on the laptop, which the authors associate with the limited computing capacity of mobile devices when executing the expensive pairing-based cryptographic operations required by ABE. The authors also observed that energy consumption is

approximately proportional to execution time. In a more constrained setting, Perazzo et al. (2021) evaluated ABE on ESP32 and RE-Mote devices and showed that ABE can significantly affect battery lifetime, especially when the number of attributes increases. For example, in the ESP32 evaluation, the estimated battery lifetime decreased from approximately 134 days without encryption to around 50–62 days when ABE was applied with 10 attributes, depending on the evaluated scheme. With 50 attributes, some schemes reduced the estimated lifetime to approximately 25 days, while the most expensive configuration depleted the battery within only a few days. These findings indicate that ABE-based IIoT architectures should consider the computational and energy profile of the device responsible for executing cryptographic operations, especially when encryption or decryption is expected to occur close to the sensor layer.

To mitigate these obstacles of high computational demand and consequent energy depletion, the work by Ziegler et al. (2019) proposed a CP-ABE architecture with outsourced partial decryption, reducing the computational burden on constrained devices. In this architecture, the client private key is split into three parts, distributed among the Key Authority, the Re-Encryption Server, and the Client. This design reduces key escrow risks, since no single actor holds the complete private key, and supports attribute revocation by updating or invalidating key parts when access permissions change.

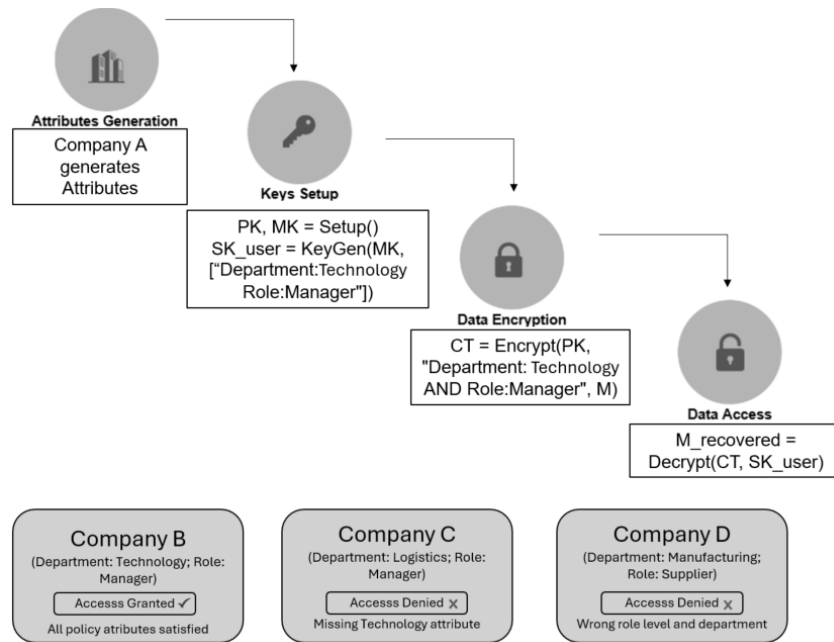
Attribute-based cryptographic methods have also proven to be a robust solution for access control in Industry 4.0 cloud environments, as highlighted in the study by Saleem et al. (2024). The authors analyze how variants of Attribute Based Encryption (ABE) — specifically ciphertext-based policy-based ABE (CP-ABE) and key-based policy-based ABE (KP-ABE) — offer superior flexibility by allowing data access to be conditioned on complex sets of attributes. The study highlights that the maturation of this technology has resulted in improved resistance to collusion and computational efficiency, with reductions in polynomial processing time and faster key generation, making it an attractive model for sharing sensitive information in decentralized industrial infrastructures.

Recent studies proposed by W3C (2025) explored the integration of ABE with Self-Sovereign Identity (SSI) as a strategy to enhance decentralized access control. SSI acts as a decentralized identity model in which each organization maintains sovereign control over digital credentials. In another practical implementation arrangement, Soltani et al. (2021) proposed that each IoT device will possess a Decentralized Identifiers (DID) registered on the Hyperledger Besu blockchain. Issuing organizations (e.g., manufacturers) will emit Verifiable Credentials (VCs) containing specific attributes (e.g., company:ABC, class:temperature, location:sectorX). When a device needs to access data, it presents a Verifiable Presentation (VP) proving its attributes,

which are validated against the blockchain registry before the corresponding ABE key is generated. These credentials allow attributes to be verified with reduced dependence on a single centralized identity provider, addressing trust issues between organizations without pre-established relationships. This approach is particularly relevant in inter-organizational IIoT environments, where no entity is fully trusted. Similarly, Ouaddah et al. (2021) proposes a privacy preservation framework that combines SSI and ABE to enforce access control in decentralized environments, demonstrating that SSI credentials can be cryptographically linked to ABE policies.

To make the operation of the cryptographic scheme used in this work clearer, Figure 2.1 illustrates the execution flow of the main methods of the *OpenABE* library, which will be used in the technology stack in this work, and whose integration will be explained in the following chapters. The example demonstrates the process from attribute generation to data access control, highlighting how keys and access policies are used to ensure that only entities with valid attributes can decrypt the information. (Zeutro LLC, 2019).

Figure 2.1: ABE Workflow.



Source: Author's Figure

As shown in Figure 2.1, the process begins with the generation of attributes by Company A, followed by the key-configuration step $Setup()$, in which the public key (PK) and the master key (MK) are created. Next, the library executes $KeyGen()$ to issue a user private key (SK) bound to a set of attributes, `Department:Technology` and `Role:Manager`. The issuer then encrypts the data using the $Encrypt()$ method, which uses the policy `Department:Technology`

AND Role:Manager. Subsequently, the consumer executes `Decrypt()`, which verifies whether the user's attributes satisfy the defined policy. In the example, only Company B has the necessary attributes and is granted access, while the other organizations are denied access due to missing or incompatible attributes. This flow demonstrates how ABE provides granular access control in multi-organizational scenarios and, in the following chapters, it will be demonstrated how the library will be made more flexible and used to support and integrate with the technological stack of this project. Table 2.1 details the main methods of the OpenABE library and their respective roles in the cryptographic process.

Table 2.1: Main OpenABE Methods and Their Functions

Method	Description
<code>Setup()</code>	Initializes the cryptographic environment and generates the system's main key pair: the public key (PK) and the master key (MK).
<code>KeyGen(MK, attributes)</code>	Generates the user's private key (SK) associated with a specific set of attributes issued by the authority.
<code>Encrypt(PK, policy, M)</code>	Encrypts the message (M) using the public key (PK) and an access policy defined over attributes, producing the ciphertext (CT). The ciphertext embeds both the AES session key (protected under ABE) and the access policy.
<code>Decrypt(CT, SK)</code>	Decrypts the ciphertext (CT) only if the attributes associated with the private key (SK) satisfy the policy defined during encryption, recovering the original message (M_recovered).

Source: Author's own work, based on Zeutro LLC (2019).

2.2.1 CP-ABE x KP-ABE

Attribute-Based Encryption (ABE) schemes are divided into two main variants: Key-Policy ABE (KP-ABE) and Ciphertext-Policy ABE (CP-ABE). In KP-ABE, the ciphertext is labeled with a set of descriptive attributes, while the decryption key contains an access policy defined by the authority. Thus, the authority determines which data a user can access by embedding

logical rules (e.g., AND, OR) into the user's private key. This model centralizes control at the key-issuing entity, which is suitable for hierarchical or single-domain systems but limits flexibility in decentralized environments.

In the case of CP-ABE, the roles are subtly reversed: the data owner is responsible for specifying the access policy and embedding it directly in the ciphertext, and each consumer holds a key associated with a set of attributes. The decryption process requires that the attributes in the consumer's key satisfy the policy defined in the ciphertext, as described in the work of Ziegler et al. (2019). This inversion of control is particularly advantageous in distributed IoT architectures, where companies can autonomously define which attributes each device will contain, so that the change only impacts data access management and permission renewal, without harming the sharing architecture.

2.2.2 CP-ABE Use Cases in Distributed Scenarios Across Organizations

In this subsection, to understand the applicability of the cryptographic method, two concrete use cases from the recent literature were analyzed. ABE was applied in distinct contexts. In both, the researchers explored the integration of ABE with decentralized systems, aiming to ensure distributed access control and data privacy across multiple domains.

1. Use Case 1 – Blockchain-based CP-ABE with Distributed KMS and Zero-Knowledge Proofs:

The study by Ren et al. (2024) proposes a data sharing model integrating CP-ABE, blockchain, and a distributed key management system (Key Management Service – KMS). The study addresses the problem of dependence on a central key-issuing authority in traditional ABE schemes, factors that, in the author's opinion, represent potential scalability limitations and single points of failure. The authors propose a distributed Key Management Service (KMS), where the CP-ABE Master Secret Key (MSK) is sharded and managed by re-encryption proxy nodes on the blockchain, eliminating the single point of failure. To overcome these limitations, the system defines two types of network nodes:

(1) re-encryption proxy nodes, responsible for managing fragments of the master secret and generating proofs of correct re-encryption (Non-Interactive Zero-Knowledge Proofs – NIZK); and (2) operator nodes, which perform off-chain transaction aggregation and publish zk-SNARKs on-chain to ensure the integrity of operations.

The scheme was tested, and after 100 concurrent transactions, the results showed an average response time of 28 seconds per node and a reduction in gas consumption of over

61%, indicating a substantial improvement in efficiency and scalability. The Ren et al. (2024) use case is relevant to this work, as it addresses similar aspects essential in interoperability scenarios, such as decentralized key management, privacy-preserving access control, and the reduction of single points of failure. However, the model does not yet explore direct communication between IIoT devices, a gap that is addressed in the work proposed in this research.

2. Use Case 2 – Multi-Authority Decentralized CP-ABE for Scalable IoT:

Wu et al. (2025) present a decentralized multi-authority CP-ABE scheme aimed at IoT applications. The work explores the smart city environment, where multiple stakeholders (e.g., traffic management, power grids, emergency services) need to share data selectively and securely.

To mitigate computational overhead on IoT devices, the model introduces outsourced computing, where high-performance Compute Nodes (CNs) assist the Controllers (CRs) in the most complex cryptographic operations. A proxy key mechanism is used to allow partial decryption off-device without exposing the original attribute key.

The scheme achieved execution times of approximately 160 ms (10-attribute key generation), 75 ms (10-attribute encryption), and almost 25 ms (final decryption), demonstrating practical efficiency. Although the key generation time is slightly longer than some competing schemes, the architecture proposes a superior balance between security, decentralization, and efficiency.

Both use cases validate the viability of CP-ABE for multi-organizational environments. In the first study, the authors Ren et al. (2024) introduce a trust layer with blockchain and zero-knowledge proofs. Meanwhile, in the second study, the authors Wu et al. (2025) offer an architecture using the cryptographic method for resource-constrained devices. This research identifies an opportunity to expand on these contributions.

2.3 Blockchain

Technological advancement and the evolution of human needs and behaviors have driven the rise of various technologies, among which blockchain stands out. Blockchain is not merely a single technique, but a technological paradigm that integrates cryptography, mathematical constructs, and distributed algorithms to address synchronization challenges in decentralized systems (LIN; LIAO, 2017). In the context of this work, the technology is known for its ability to store transactions in an immutable, sequential, and distributed manner. Its fundamental

principles are characteristics that differentiate it and confer significant benefits compared to traditional computing systems, namely: Peer-to-Peer System, Cryptographic Mechanisms, Distributed Consensus Algorithm, Open Source, and Economic Incentives.

2.3.1 Blockchain Pillars

The architecture of blockchain technology is based on five pillars that, together, make it possible to implement a reliable decentralized system. These are:

1. **Peer-to-Peer System:** Blockchain operates on a peer-to-peer network topology, which ensures that the nodes involved can see and participate in established communication and transaction validation. Unlike traditional networks, and as described by Tschorsch & Scheuermann (2016), this topology eliminates the need for a central server to control communication, increasing the system's failure resistance and availability. For example, Zeydan et al. (2023) note that blockchain systems are built on P2P (Peer-to-Peer) networks, so that nodes can propagate transactions among themselves and organize them in a distributed manner.
2. **Cryptographic Mechanisms:** In the context of this work, it will be necessary for the information shared between different companies to remain intact and for executed transactions and granted access permissions to be immutable. Considering this, blockchain is particularly convenient in this architecture because, due to its cryptographic mechanisms, such as hash functions and Merkle trees, the transaction history remains protected. As Tripathi et al. (2023) pointed out in their recent studies, public key cryptography and hash functions ensure the immutability and verifiability of records.
3. **Distributed Consensus Algorithm:** This is the main mechanism of the technology and is responsible for ensuring that decentralized nodes can converge on an understanding of the state of information. This feature is crucial in networks without centralized authorities, allowing nodes to validate and include blocks in a coordinated and reliable manner. Consensus algorithms constitute a critical component of blockchain systems, as they enable distributed nodes to agree on the state of the ledger (HUSSEIN et al., 2023). Algorithms such as Proof of Work (PoW) and Proof of Stake (PoS) were designed to ensure that honest participants agree on the order and validity of transactions, even in the eventual presence of malicious nodes (CACHIN; VUKOLIC, 2017).
4. **Open-source:** Blockchain projects usually rely heavily on collaboration from the technical community, as they are generally made available as open-source software, and this

open-source nature mitigates trust in the technology's protocols over time.

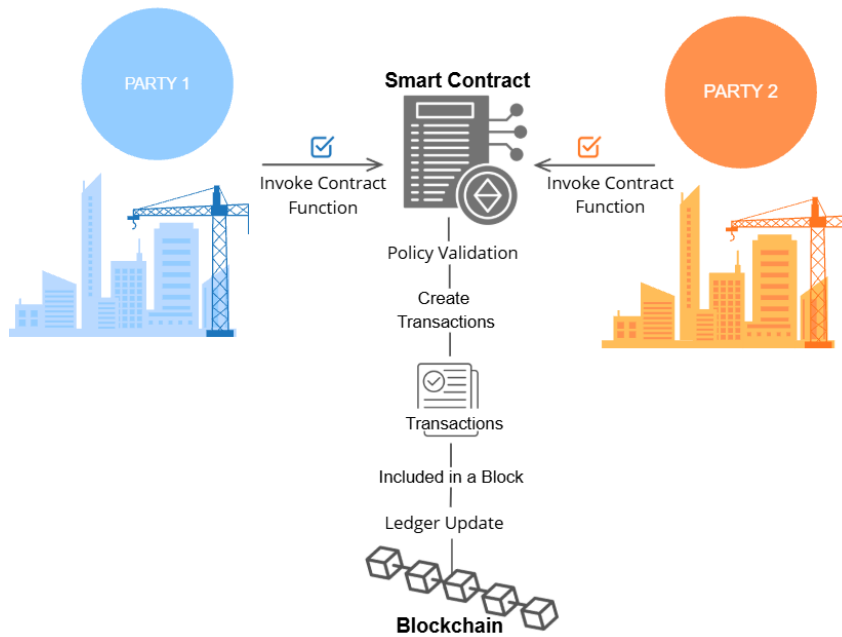
5. **Economic Incentives:** To ensure blockchain stability, many protocols, especially public ones, adopt a system of economic rewards for nodes that actively participate in the network, dedicating resources to validating transactions and creating new blocks. In this sense, Ding et al. (2021) investigated how incentives attract corporations and IoT devices to contribute computing power to blockchain platforms, concluding in their studies that the presence of economic incentives aligns individual interests with the collective functioning of the network. Incentives include, for example, rewards to entities that validate blocks, transaction fees, or token-economy systems.

Synergistically, the aforementioned pillars form the technical and economic basis that allows blockchain to offer properties such as immutability, auditability, fault tolerance, and distributed trust. These characteristics are necessary for the proposed architecture and will serve, structurally, as the theoretical foundation for the following sub-items of the dissertation.

2.3.2 Smart Contracts

This technology can be described as computer programs that run automatically on the blockchain when certain conditions defined in their code are met. This involves the possibility of functions being executed like contractual clauses, but without the need for direct human intervention. As reviewed by Khan et al. (2021), contracts are protocols designed to automatically verify and enforce agreements and rules between parties who do not trust each other.

In their importance for distributed systems, smart contracts directly link business logic to the Distributed Ledger Technology (DLT) infrastructure layer, promoting immutability and auditability. Regarding their operation, the process typically involves: (1) programming the contract and its deployment on the blockchain; (2) invoking the contract through a network transaction that satisfies a conditional structure; (3) automatically executing the contract's functions, which represent the contractual clauses, and recording the result on the blockchain; and (4) termination or re-invocation, according to the outlined flow. This operational cycle is illustrated in Figure 2.2 and demonstrates how different parties interact, through transactions, with the smart contract.

Figure 2.2: Smart Contract Operation Flow.

Source: Author's Figure

The smart contract acts as a component for registering devices and attributes, granting permissions, and managing the distribution of cryptographic keys among the parties involved. Specifically, the contract is responsible for the data access logic, depending on the established attributes and policies. Its importance lies in its ability to create transparent and tamper-proof processes, characterized by being immutable after deployment, having deterministic execution, and operating in a decentralized environment (ALHARBY; MOORSEL, 2017).

2.3.3 Hyperledger Besu

For this work, the existence of different types of Blockchain was considered, each containing different implementations and properties that impact the performance and governance of the solution. Among the available options, Hyperledger Besu was chosen because it stands out as a platform compatible with Ethereum, designed to operate in both public networks and private, permissioned environments, which made it particularly suitable.

Hyperledger Besu is an open-source Ethereum client, developed in Java under the Apache license, that implements the components of an EVM client: state storage, P2P networking, block synchronization, and consensus mechanisms such as Proof-of-Authority (FAN et al., 2022). Its modular architecture supports integration with industrial and enterprise systems, and this interesting flexibility makes Besu especially suitable for scenarios where control over

participant identity and transaction traceability are important security and compliance requirements (CAPOCASALE et al., 2023).

In this research proposal, Hyperledger Besu serves as the immutable ledger and smart contract execution layer that orchestrates communication between IIoT entities. Its compatibility with the Ethereum model allows the use of smart contracts to record events and distribute cryptographic keys. Operating in permissioned mode, Besu enables distributed governance between companies, reducing dependence on a central authority and ensuring transparency in interactions. The following chapter, “Methodology”, describes the procedures adopted in this research and architecture, covering node configuration, integration with the MQTT broker, and the instrumentation of the performance tests that support the experimental validation of the proposal.

Chapter 3

METHODOLOGY

This chapter presents the methodological approach adopted in this research, describing the research paradigm, the systematic literature review process, the design and implementation of the proposed artifact, and the evaluation strategy adopted to validate the solution. These procedures were defined to ensure coherence between the identified research problem, the theoretical foundation, and the proposed technological artifact.

3.1 Design Science Research

This research adopts the Design Science Research (DSR) paradigm, which focuses on the design, development, and evaluation of technological artifacts intended to solve identified problems in specific application domains (ENGSTRÖM et al., 2020). Unlike purely descriptive research approaches, DSR emphasizes the construction and validation of artifacts capable of addressing practical challenges while also contributing to scientific knowledge (HEVNER et al., 2004).

The methodological structure proposed by Peffers et al. (2007) was adopted as the main research framework. The process involves problem identification, definition of solution objectives, artifact design and development, demonstration, evaluation, and communication of results. This approach is appropriate for this research because the study is not limited to analyzing privacy, access governance, and cross-organizational coordination challenges in IIoT environments, but also proposes and evaluates a functional architecture intended to mitigate these limitations.

The initial stage of the research consisted of conducting a Systematic Literature Review (SLR) to identify the main challenges related to cross-organizational interoperability, distributed access control, and data privacy in multi-organizational IIoT environments. The review also investigated the use of blockchain technologies, MQTT-based communication architectu-

res, and Attribute-Based Encryption (ABE) mechanisms in distributed industrial scenarios. The SLR was not treated as an isolated activity, but rather as part of the rigor cycle of DSR, providing the theoretical basis required to guide architectural decisions and identify limitations in existing approaches.

Based on the identified challenges, the main artifact developed in this research consists of a secure cross-organizational architecture for granular and governed data exchange in the IIoT. The proposed solution integrates a publish/subscribe communication layer implemented with MQTT, a permissioned blockchain network implemented with Hyperledger Besu, and attribute-based encryption mechanisms to provide granular access control over the shared data.

The solution was implemented in a controlled local environment using real IIoT humidity and temperature sensors to publish data via MQTT topics. Attribute-based encryption is applied to enforce access restrictions during the data sharing process. The evaluation phase considers metrics related to encryption and decryption latency, transaction confirmation time on the blockchain, computational overhead, and system behavior under different message sizes and attribute configurations.

Table 3.1 summarizes how the stages proposed by Peffers et al. (2007) are associated with the activities performed throughout this dissertation.

Table 3.1: Association between DSR stages and dissertation activities

DSR Stage		Application in this Research
Problem Identification	Identification	Identification of cross-organizational interoperability, privacy, and distributed access control limitations in multi-organizational IIoT environments, based on the problem statement and literature review presented in Chapter 4.
Define Objectives of a Solution	Define Objectives of a Solution	Definition of the research objectives and architectural requirements, including data sharing and privacy between different companies using blockchain and Attribute-Based Encryption mechanisms.
Design and Development	Design and Development	Design and implementation of the proposed interoperability architecture integrating MQTT, blockchain, and an Attribute-Based Encryption module, as described in Chapter 5.
Demonstration	Demonstration	Deployment of the experimental environment and execution of functional integration tests involving real IIoT devices, MQTT communication, Blockchain registration, and protected data exchange.
Evaluation	Evaluation	Experimental evaluation of the proposed solution considering encryption and decryption latency, and behavior under different attribute configurations and message sizes, as presented in Chapter 6.
Communication	Communication	Documentation and dissemination of the research results through this dissertation, scientific publications, diagrams, implementation artifacts, and public repositories.

Source: Author's own work, based on Peffers et al. (2006).

According to Gregor & Hevner (2015), DSR plays an important role in computing research by enabling the systematic development and evaluation of technological artifacts. In this

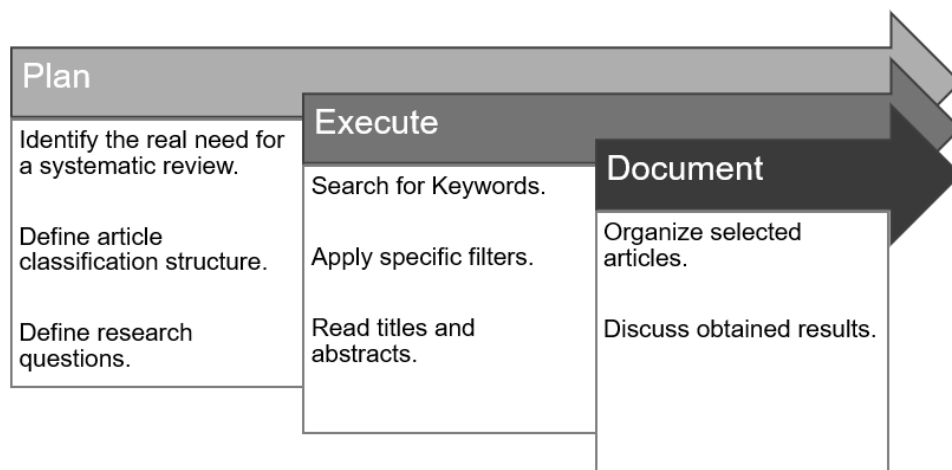
research, the generated results are documented through scientific publications, while the implementation artifacts are made available in a public repository to support reproducibility and future extensions.

3.2 Systematic Literature Review

The formulation of the research problem and the definition of the objectives for this work were structured through the execution of a **Systematic Literature Review (SLR)**. In a standardized and structured manner, the review was based on the guidelines proposed by Kitchenham & Brereton (2013). As proposed by the authors, three stages were adopted. In the planning phase, initial parameters were defined, such as research questions, inclusion and exclusion criteria, and search strategies in scientific databases. Subsequently, in the execution phase, several searches were carried out in the previously stipulated databases, followed by the selection of studies and data extraction, guided by the inclusion and exclusion criteria. Finally, the documentation phase consisted of identifying research gaps and synthesizing the results.

This structured process is illustrated, with adaptations, in Figure 3.1.

Figure 3.1: Steps of the process adopted for the Literature Review



Source: Adapted from (KITCHENHAM; BRERETON, 2013)

A systematic literature review (SLR) was conducted to map existing solutions and identify research gaps related to the proposed topic. Following the adopted protocol, keywords were defined to ensure broad coverage of scientific publications. The Mendeley Reference Manager tool was used to assist in cataloging by section, managing and organizing the researched articles and their respective databases and references.

3.3 Experimental Plan

The implementation of the proposed architecture is divided into two main stages. The first stage consists of testing the layer containing the cryptographic mechanisms, with the aim of evaluating the impact of different access policy complexities and payload sizes. These tests measure both processing overhead, based on encryption and decryption times, and spatial overhead, based on the increase in payload size after ABE protection. The second stage involves a functional validation of the integrated environment, simulating the flow from data generation on IIoT devices to registration on the blockchain. This stage focused on verifying the interoperability between the MQTT broker, the Python orchestrator, and the Hyperledger Besu blockchain. The integrated validation focused on transaction confirmation success and the consistency of the audit records generated by the smart contract.

Chapter 4

SYSTEMATIC LITERATURE REVIEW

Following the methodological structure presented in Section 3.2, this chapter presents the results of the Systematic Literature Review (SLR). The objective of the review was to identify existing approaches, technological trends, and related research gaps. The results provide the theoretical basis for the architecture proposed in this research, as well as assisting in the decision-making process regarding the chosen technologies.

Initially, the criteria adopted for study selection and the scientific databases consulted are described. Subsequently, the selected works are analyzed with emphasis on architectural approaches, cryptographic mechanisms, interoperability strategies, and identified technological limitations relevant to secure cross-organizational data exchange.

4.1 Research Questions

As described by Kitchenham & Brereton (2013), research questions are defined during the planning phase of a Systematic Literature Review and establish the scope and direction of the investigation. The research questions were formulated to explore the integration of distributed communication mechanisms, blockchain-based auditability, and attribute-based encryption in IIoT environments. The questions are presented in Table 4.1.

Table 4.1: Research Questions

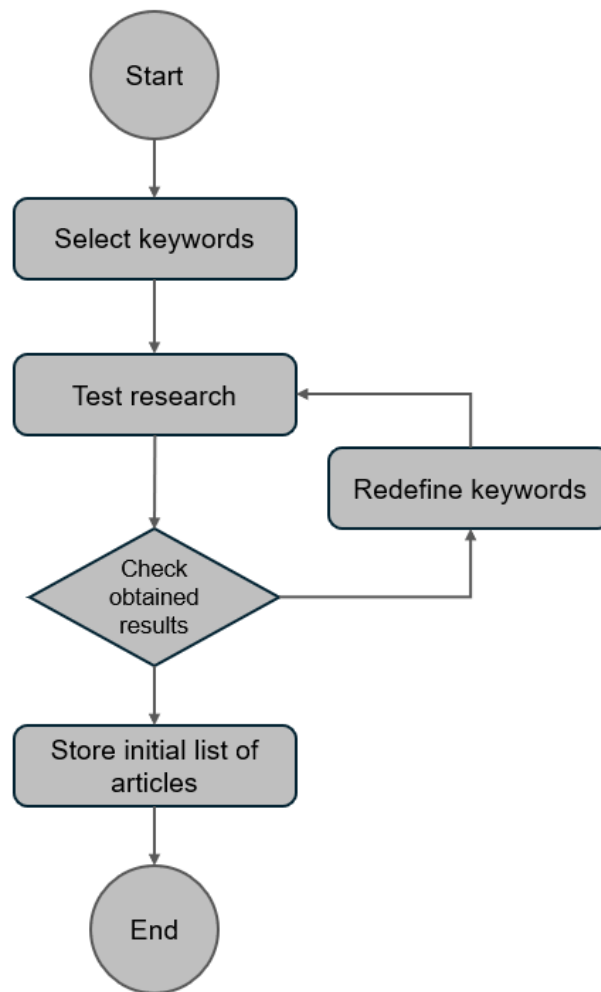
Question	Description
Q1	How can Attribute-Based Encryption (ABE) enhance access control and privacy in Industrial IoT systems integrated with blockchain platforms?
Q2	How can blockchain technologies be integrated with existing IoT systems to support secure and auditable data exchange in multi-organizational industrial environments?
Q3	What is the impact of combining blockchain technology with publish/subscribe communication architectures on the performance and efficiency of IIoT systems?

Source: Author’s own work.

With the research questions defined, four scientific databases were selected for the literature search:

- ACM Digital Library
- IEEE Digital Library
- ScienceDirect
- Springer Link

According to Centre for Reviews and Dissemination (2024), the definition of research questions and search criteria at the planning stage contributes to reducing bias and ensuring transparency throughout the review process. The identification of search terms followed the approach proposed by Paul et al. (2021), in which keywords are extracted from the research questions and iteratively refined. The search protocol was structured based on the strategy described by Unterkalmsteiner et al. (2012), illustrated in Figure 4.1.

Figure 4.1: Research strategy

Source: Adapted from Unterkalmsteiner et al. (2012)

4.2 Literature Search Strategies

The search strategy was developed according to the guidelines of Kitchenham & Brereton (2013), focusing on interoperability, blockchain integration, attribute-based encryption, and distributed communication architectures in IIoT environments.

The search string was structured as follows:

(“Interoperability in IoT Environments”AND “Blockchain”) OR (“Interoperability in IoT Environments”AND “Attribute-Based Encryption”) OR (“Interoperability in IoT Environments”AND “Publish/Subscribe Architecture”) OR (“Interoperability in IoT Environments”AND “Message-Oriented Architecture”)

Aiming to reduce the chances of relevant works being overlooked in the investigations, the terms were selected to reflect the breadth of the concepts in this research, and not just specific software implementations. It should also be noted that specific adaptations were applied to each database, as there are differences in the search mechanisms between IEEE Xplore, the ACM Digital Library, ScienceDirect, and Springer Link. A filter with a time range covering the last six years was applied to ensure relevance to technological developments, seeking current works.

In the analysis of the studies used, the aim was to find architectural patterns related to secure data exchange, similar cryptographic approaches, always considering the performance of the results presented. Special attention was given to solutions that address the secure exchange of data between independent organizations and the application of decentralized access control policies.

4.3 Inclusion Criteria

To ensure the relevance and quality of the studies analyzed, the inclusion criteria presented in Table 4.2 were adopted. These criteria defined the scope of the search and selection process.

Table 4.2: Inclusion Criteria

Criterion	Description
Databases	IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink.
Search Terms	“Interoperability in IIoT Environments”, “Blockchain”, “Attribute-Based Encryption”, “Publish/Subscribe Architecture”, “Message-Oriented Architecture”.
Methodological Quality	Studies with clearly defined and described methodologies, including implementation details, evaluation metrics, and experimental validation where applicable.
Relevance	Articles addressing security, privacy, decentralized governance, and interoperability in multi-organizational IIoT contexts.
Language	Articles published in English.

Source: Author’s own work.

4.4 Exclusion Criteria

To maintain the rigor and feasibility of this project, exclusion criteria were defined to eliminate studies that did not meet the established methodological requirements. The selection process involved an initial screening based on titles and abstracts, followed by full-text assessment of potentially relevant studies. The exclusion criteria are presented in Table 4.3.

Table 4.3: Exclusion Criteria

Criterion	Description
Lack of Integration Focus	Studies addressing only isolated aspects (e.g., solely blockchain or solely ABE) without considering cross-organizational interoperability, secure data exchange, or integrated architectures.
Publication Type	Opinion papers, abstracts without full text, slide presentations, or non-peer-reviewed technical reports.
Insufficient Methodological Detail	Studies lacking adequate description of methods, experimental setup, or validation procedures.
Duplication	Duplicate publications, with preference given to the most complete and recent version.
Restricted Access	Studies without full-text availability through institutional access.
Publication Period	Articles published more than six years prior to the search period.

Source: Author's own work.

The study selection process began with screening titles and reading abstracts to exclude works clearly irrelevant to this research. Subsequently, the full text was read for the remaining studies, focusing more on the methodologies adopted and the results obtained. For the selected articles, an analysis of the experimental environments and findings involving data was performed. For these, and using the Mendeley Reference Manager tool, the title, authors, year of publication, main results, limitations, and relevance to the research objectives were recorded.

The quantitative summary of the review was consolidated based on search records obtained from the consulted databases and the references organized during the review process. The initial search returned 87 records across IEEE Xplore, ACM Digital Library, ScienceDirect, and SpringerLink. After applying the inclusion and exclusion criteria defined in the review protocol, 23 studies were selected for qualitative analysis.

The remaining records were excluded during the screening and eligibility assessment stages

for one or more of the following reasons: lack of alignment with the research scope, absence of an integrated architectural perspective, insufficient methodological or experimental details, duplication, restricted access to the full text, publication type incompatible with the review protocol, or publication outside the adopted time period.

4.5 Results

This section presents the conclusions obtained from the review. The selected studies were analyzed comparatively, mainly considering the proposed architectures, cryptographic mechanisms used, interoperability strategies adopted, and experimental validation approaches. The objective was to ensure that the included studies represented the state of the art on the subject and provided a basis for the implementation of the proposal in this work. To facilitate the analysis, the results are organized into two complementary perspectives: (i) specific approaches to privacy and security in IIoT environments, and (ii) interoperability architectures designed for data exchange between organizations. This classification allows for a more defined identification of the technological trends, limitations, and gaps found.

4.5.1 Specific approaches to privacy and security

Among the various challenges involved in IIoT ecosystems, privacy and security become particularly complex when data must be exchanged across organizational boundaries. In such scenarios, technical connectivity alone is insufficient, since organizations also need mechanisms to preserve confidentiality, enforce access policies, and maintain auditability over shared information. From this perspective, the combination of technologies such as blockchain, publish/subscribe architectures, and ABE emerges as a promising approach to support secure and governed data exchange.

This literature review investigated relevant studies that explore nuances of this approach, analyzing proposed architectures, seeking to understand existing gaps, and how they can be improved. The application of blockchain technology in different sectors has demonstrated its potential to support security, auditability, and controlled data exchange in environments involving sensitive information. In this context, Yu & Mu (2024) proposed a multilayer blockchain system with ABE for data exchange in smart agriculture, demonstrating the potential of this combination in different sectors. In this work, given its characteristics of immutability and decentralization, the IPFS (Interplanetary File System) protocol was used, along with the HTLC (Hash Time-Locked Contracts) technique, ensuring that transactions were completed within a

predefined time window; otherwise, the smart contract automatically unlocked the assets according to the expiration condition, guaranteeing the integrity and authenticity of the data. Despite presenting promising results, large-scale practical application still requires further study, as the authors concluded, highlighting the need for investigation into decentralized data collection and automation of contracts.

The importance of testing decentralized models in data interoperability solutions in IoT environments was highlighted by Mishra et al. (2023), who presented a decentralized access control system based on blockchain and IPFS, implemented in a test network using Ethereum. Despite the challenges of complexity and the need for robust computational resources, analyzing the results obtained, a significant reduction in resource consumption and an improvement in the efficiency of processing new records was observed, where the time to create access policies increased linearly with the number of simultaneous requests, but remained efficient. Smart contracts in Solidity automated access control, and the solution was evaluated in terms of execution time and gas consumption.

One of the biggest challenges in interoperability solutions between IIoT devices is ensuring the security and privacy of transmitted data. Khan et al. (2023) addressed this challenge using Hyperledger Fabric and NuCypher threshold re-encryption, showing a reduction in resource consumption and a significant improvement in the privacy of data stored on the blockchain. Integration with healthcare systems was performed using RESTful APIs. Although the study provides a robust solution for IIoT data security, specifically in e-healthcare environments using blockchain, no predictions or scalability tests in real-world scenarios were presented.

Efficient access policy management is fundamental to ensuring data security and traceability. In their research, Guo et al. (2019) proposed a multi-authority ABE scheme for IoT device interoperability, ensuring data integrity and traceability. Tested in a telemedicine environment, the solution ensured efficient access policy management and key updates, using Ethereum blockchain and Solidity smart contracts. Despite the flexibility and security demonstrated with the proposed ABE scheme, the complexity of managing multiple authorities and the need for a robust infrastructure were identified as limitations.

Throughout this literature review, it was noted that, despite the availability of several studies on the subject, little focus is given to the specific privacy challenges associated with the integrated blockchain-IIoT architecture. From this perspective, Gugueoth et al. (2023) conducted a comprehensive review on the use of decentralized blockchain techniques for privacy in environments of this nature. Using the Hyperledger Fabric platform to exemplify practical solutions, the scalability of the solutions and the need for more empirical studies were identified as areas

that require attention, in addition to raising the need to analyze the cost-benefit of large-scale implementations, compared to the expected benefits.

Recent studies have investigated hybrid architectures that combine blockchain and messaging systems to support access control, asynchronous communication, and secure data exchange in IoT ecosystems. Karankar & Seth (2025) proposed an access control system for IoT that integrates a permissioned blockchain with a message queuing system (RabbitMQ) to decouple transaction validation from communication between devices. Subtly different from the proposal in this dissertation, in their architecture, the authors proposed that smart contracts implement on-chain attribute-based access control (ABAC), while the message broker handles asynchronous communication between devices and the blockchain layer. The experimental evaluation analyzed the execution time and overhead of policy management, including measurements of the time to create access policies as the number of policies increased. The authors identified throughput limitations and congestion in the blockchain component when operating independently, highlighting the need for adaptations to achieve scalability improvements to accommodate large volumes of distributed devices.

Also in this context, Mahalle et al. (2025) proposed an interoperability framework structured around standardized communication protocols, semantic data models, and a modular architecture. The framework incorporates a uniform data representation to enable semantic interoperability between devices using different protocols. Experimental results reported improvements in data transmission accuracy, reduced packet loss, and increased interoperability compliance. The authors conducted an evaluation in a local environment and presented positive performance metrics, although they did not perform larger-scale tests or decentralized integration of the architecture.

From an industrial perspective, Afrin et al. (2025) conducted a review of IIoT implementations in different industry sectors, examining configurations involving edge, fog, and cloud layers. The study identified interoperability gaps arising from the management of distributed data and integration between different domains. While the article synthesizes existing implementation strategies and discusses hybrid and layered architectures, it emphasizes that large-scale validation and cross-domain integration remain unresolved challenges, particularly in scenarios involving legacy systems.

Privacy preservation in industrial IoT was also addressed by Mahendran et al. (2025) through the PRISM-IIoT framework. The authors integrated blockchain, lightweight cryptography (LiCrypt), and cascading deep learning (CasDL) for intrusion detection. The architecture adopts a consortium blockchain model to record access and usage events, enabling data

auditability and integrity. The experimental evaluation measured accuracy, latency, scalability, power consumption, and adaptability with increasing numbers of devices. Promisingly, the results indicated maintained scalability with 500 devices, an average blockchain transaction latency of 22 ms, and a throughput of 120 transactions per second. However, the authors acknowledge interoperability challenges when integrating the framework with legacy industrial systems that lack support for modern cryptographic protocols or blockchain communication layers.

Data security and privacy represent a crucial challenge in IIoT environments, as highlighted by Peter et al. (2022), who point to the lack of infrastructure and skilled labor as obstacles to the implementation of IIoT solutions in emerging economies. The main barriers include a lack of adequate infrastructure, high implementation costs, resistance to change, and concerns about data security. Although it mentions tools such as ThingWorx and Predix, the article does not analyze in detail the architecture, protocols, and usability testing with these tools. Alsolami et al. (2021) proposed an interoperable access framework that improves resource utilization in heterogeneous IoT applications, using transfer learning to replace unavailable services. Tests conducted using TensorFlow for machine learning and Hyperledger Fabric for blockchain highlighted resource management and adaptation to different usage contexts as identified challenges.

The security and privacy of shared data are also essential, especially in sensitive contexts. In this sense, Chen et al. (2020) proposed GovChain, a framework that integrates ABE and Blockchain to share data in e-government systems. GovChain uses a blockchain-based trusted identity authentication architecture and stores encrypted documents in an off-chain system (IPFS), while document indexes are stored on-chain. The framework was designed to minimize the computing cost of ABE, allowing users to access encrypted data according to predefined attributes. The authors presented and implemented the solution on an Ethereum network, and the results demonstrate that GovChain proved effective, ensuring privacy and granular access control for data sharing.

Addressing the challenge of integrating heterogeneous systems from different administrative levels, the authors Loss et al. (2023) proposed a blockchain strategy for interoperability in multi-level public security scenarios. The case study presented by the authors used the scenario of Rio Grande do Norte, in Brazil, illustrating how the Hyperledger Fabric blockchain could be used to connect systems at the municipal, regional, and national levels. The solution offers a model for building an interoperable and secure system, allowing collaboration between different public security agencies at different hierarchical levels. It is noted that, although the research mentions the need to test scalability, it does not explore what behaviors would be observed in

these scenarios, with a high number of systems and users.

4.5.2 Interoperability Architectures for Secure Data Sharing

Different IoT networks and systems can belong to various organizations or entities, whether they collaborate with each other or not, while maintaining partial or complete autonomy over the data produced by their devices. Integrating these different devices for this granular and secure data distribution is particularly useful in scenarios where multiple organizations need to work together. In this context, recent work conducted by Sychowiec & Zielinski (2023) explored the integration of Hyperledger Fabric with Apache Kafka for data distribution in federated IoT environments. The Kafka Streams API was used for processing the data stream generated in real time. Tests were performed in a federated environment on the AWS cloud and showed a reduction in latency and an improvement in security against malicious attacks. The integration of Hyperledger Fabric, with contracts using the Go and Node.js languages, demonstrated the viability of an efficient architecture for federated data exchange. One important issue that remains unresolved, however, is the acquisition, analysis, and integration of the enormous amounts of data generated from decentralized environments.

An interoperability gateway for microservices-based systems using Kafka and RabbitMQ was proposed by Bayramcavus et al. (2021), demonstrating effectiveness in bidirectional communication between different middleware systems. In the study, the researchers conducted practical tests involving two different systems, each with a different middleware (RabbitMQ and Kafka) in a controlled environment, using Java APIs and the Spring Boot framework for portal implementation. To a certain extent, communication delays between the components were similar, but increased significantly under very high loads. The use of Apache Kafka proved effective in supporting communication between heterogeneous systems, standing out as a promising alternative for distributed integration scenarios.

The need for large-scale performance analysis and evaluation of interoperable standards was highlighted by Singamaneni et al. (2024), who conducted a comparative analysis of ABE standards, evaluating various standards in cloud, IoT, and blockchain environments. Using platforms such as AWS for the cloud and Ethereum for blockchain, the tests compared the performance of different ABE standards when integrated with IoT devices in industrial environments. Tools such as OpenSSL were used for cryptography and performance testing. It was concluded that certain models offer high expressiveness and flexibility in creating access policies, but with greater computational complexity. Detailed case studies demonstrating the practical implementation of ABE standards in industrial scenarios would be beneficial.

The need for systems to handle the large amount of data in IoT, coupled with the need to ensure the privacy and traceability of this data, drives research into integrated solutions that combine blockchain and tools that support this volume of data, such as Apache Kafka. In view of this, Cui et al. (2019) proposed a blockchain-based IoT data management platform, focusing on traceability. The platform stands out for including management hubs to mediate requests from devices with limited resources, and used Hyperledger Fabric and an Apache Kafka cluster in its implementation. As the authors highlighted, Kafka ensured the platform's performance and scalability.

In IIoT environments, the implementation of Kafka-based microservice aggregation mechanisms reduces latency and increases reusability, as demonstrated by Rath et al. (2023). The solution was implemented using Kubernetes for container orchestration and Apache Kafka for communication. Experimental results showed that the developed architecture not only improves integration and interoperability, but also allowed for easy device updates, although management complexity was a significant limitation.

In their work, Fernandez-Gago et al. (2024) proposed a framework for trust interoperability in IIoT environments, focusing on the interaction between distinct trust models that can coexist in the same scenario. The framework was developed in Java and validated by the authors through a Field Service Team scenario. The article presents an implementation for a specific use case, but does not detail the complexity of large-scale implementation, considering the heterogeneity of systems and devices in an IIoT scenario.

In addition to investigations focused on implementation, Kim & Vong (2025) conducted a literature review of cross-organizational blockchain-based secure network architectures, including IIoT domains. Their structured taxonomy categorizes interoperability architectures by application field and network type, highlighting blockchain as a reliable and scalable foundation for decentralized interaction between large networks of devices. Nevertheless, the study identifies scalability, compliance, and security vulnerabilities as recurring limitations in real-world deployments, reinforcing the need for architectures that balance performance, regulatory compliance, and the ability to manage the operational complexity of the architecture.

Regarding the perspective of granular access control in interoperability solutions for IoT devices, Shakarami et al. (2022) proposed a blockchain-based architecture for smart homes. The solution was tested using Ethereum's Ropsten network and obtained promising results in terms of processing time and gas consumption. The possibility of achieving such granular control, considering aspects of each device, organization, and user, allows for the proposal of traceability strategies, something especially useful in supply chains and production scenarios.

In this context, Dasaklis et al. (2019), using IoT and blockchain, proposed a framework capable of guaranteeing end-to-end traceability, focusing on the unique characteristics of each product in a food supply chain. The methodology uses smart contracts to manage adaptable traceability according to user needs. Although it has faced challenges related to management of large volumes of data, the experiments carried out have shown that the average cost of a transaction is on the order of milliseconds, such as the deployment of the contract constructor, and therefore the solution proves to be viable for real-time traceability.

Studies conducted by Zhang et al. (2021) proposed a KP-ABE (Key Policy Attribute Based Encryption) scheme for blockchain-based IoT access management. The solution was tested using Ethereum and smart contracts in Solidity. The authors compared the traditional authentication format in IoT environments with the proposed attribute-based authentication and management. The analysis of the selected studies reveals that the combination of blockchain and ABE presents great potential to support flexible access policies and refined access control in IoT.

Analysis of selected studies indicates that the integration of blockchain technology, attribute-based cryptography, and distributed communication architectures currently provides a consistent foundation for secure interoperability architectures in IIoT environments. While hybrid blockchain and messaging architectures demonstrate performance improvements in controlled environments, scalability bottlenecks, integration issues with legacy systems, and computational overhead remain critical barriers.

4.6 Discussion

Research indicates that combining blockchain technology with attribute-based architecture can increase privacy levels and support data sharing in industrial IoT environments. The potential of blockchain technology as a solution for governing sharing rules and transaction transparency is also highlighted. Despite the promising results observed, some important limitations that must be considered are also evident.

4.6.1 Study Limitations

Several distinct limitations were observed, ranging from technical challenges to obstacles in the flow of business between companies. However, the most prominent challenges were those related to technology stack integration, decentralized data management, access policy

management, and the complexity of large-scale implementation. These limitations represent barriers that require further investigation to improve the effectiveness of currently developed solutions. Furthermore, most studies were conducted in controlled or simulated environments, which may not fully reflect conditions in real-world scenarios. For example, issues related to latency, network reliability, and compatibility between different technologies still need to be explored in greater depth. Table 4.4 summarizes the limitations identified in each study analyzed, providing a clear view of the critical points that still need to be addressed.

Table 4.4: Systematic Literature Review Results: Blockchain and ABE Applications in IoT and Data Privacy

Authors (Year)	Title	Identified Limitations	Advances / Positive Results
Yu and Mu (2024)	ABE-Based Post-quantum Cross-Blockchain Data Exchange Approach for Smart Agriculture	Requires further study on decentralized data collection and smart contract automation	Implementation of post-quantum ABE with support for cross-blockchain data exchange
Sychowiec and Zielinski (2023)	An Experimental Framework for Secure and Reliable Data Streams Distribution in Federated IoT Environments	Challenges in acquisition, analysis, and integration of large volumes of data generated in decentralized environments	Experimental tests demonstrated latency reduction and security improvements in federated IoT environments
Guo et al. (2019)	Flexible and Efficient Blockchain-Based ABE Scheme With Multi-Authority for Medical On-Demand in Telemedicine System	Deployment complexity and requirement for robust computational resources	Efficient access control for telemedicine, with enhanced security through multi-authority ABE integrated with blockchain

Authors (Year)	Title	Identified Limitations	Advances / Positive Results
Khan et al. (2023)	Data Security in Healthcare Industrial Internet of Things With Blockchain	Lack of scalability evaluation under realistic, large-scale operational conditions	Integration of blockchain mechanisms to increase privacy and security in healthcare IIoT data flows
Zhang et al. (2021)	Secure ABE Scheme for Access Management in Blockchain-Based IoT	Complexity of access management and requirements for secure key/attribute administration	ABE-based access management with security and traceability for IoT environments integrated with blockchain
Gugueoth et al. (2023)	A Review of IoT Security and Privacy Using Decentralized Blockchain Techniques	Need for more empirical studies and cost-benefit analysis in large-scale deployments	Consolidated review of decentralized blockchain techniques for IoT security and privacy, outlining open research gaps
Bayramcavus et al. (2021)	Interoperability of Microservice-Based Systems	Communication delays between components under very high loads	Practical demonstration of microservice interoperability using publisher-subscriber architectures
Singamaneni et al. (2024)	Decoding the Future: Exploring and Comparing ABE Standards for Cloud, IoT, and Blockchain Security Applications	Higher computational overhead for certain ABE models and policy configurations	Comparative analysis of ABE standards, highlighting flexibility and applicability across cloud, IoT, and blockchain scenarios
Cui et al. (2019)	IoT Data Management and Lineage Traceability: A Blockchain-based Solution	Requires management hubs and stronger performance/scalability guarantees for large deployments	Proposal of a data management and lineage traceability platform using Hyperledger Fabric and Kafka

Authors (Year)	Title	Identified Limitations	Advances / Positive Results
Peter et al. (2022)	Industrial Internet of Things (IIoT): Opportunities, Challenges, and Requirements in Manufacturing Businesses in Emerging Economies	Infrastructure limitations, high implementation costs, and data security concerns	Identifying the challenges and requirements of the IIoT in emerging economies, including barriers related to data sharing.
Alsolami et al. (2021)	Interoperable Access Framework for Internet of Things Backhauled Heterogeneous Applications	Resource management constraints and adaptation challenges across heterogeneous IoT environments	Interoperable IoT access framework with mechanisms aimed at improving resilience and resource utilization
Rath et al. (2023)	Microservice Based Scalable IoT Architecture for Device Interoperability	Complexity of coordinating microservices and managing interoperability across heterogeneous IoT components at scale	Scalable microservice-based IoT architecture supporting device interoperability, evaluated through experimental analysis
Fernandez-Gago et al. (2024)	Trust Interoperability in the Internet of Things	Complexity of large-scale implementation and trust model coordination across domains	Trust interoperability framework enabling coexistence of heterogeneous trust models in IoT ecosystems
Chen et al. (2020)	Blockchain/ABE-based Fusion Solution for E-Government Data Sharing and Privacy Protection	High computational cost of ABE and dependency on robust infrastructure	Privacy-preserving e-government data sharing with attribute-based policies enforced via blockchain integration

Authors (Year)	Title	Identified Limitations	Advances / Positive Results
Loss et al. (2023)	Blockchain Strategy for Multi-level Interoperability in Public Safety Scenario	Complexity of managing multiple operational scenarios and administrative layers	Blockchain-based strategy supporting interoperability across multiple organizational levels in public safety
Shakarami et al. (2022)	Blockchain-Based Administration of Access in Smart Home IoT	Unsatisfactory response time for administrative tasks in some operational models	Demonstrated feasibility of granular and secure access administration for smart home IoT using blockchain
Dasaklis et al. (2019)	Defining Granularity Levels for Supply Chain Traceability Based on IoT and Blockchain	Scalability challenges under high transaction volume and short processing windows	Framework for end-to-end traceability with granularity levels, leveraging IoT and blockchain to enhance integrity
Mishra et al. (2023)	Blockchain Powered IoT Access Control Model for Secure Data Sharing and Management: Performance Analysis	Limited coverage of interoperability/privacy dimensions beyond access control performance	Access control model proposal with performance analysis; emphasizes secure data sharing and management
Mahalle et al. (2025)	Building a Unified Framework to Standardize and Improve Interoperability in IoT Systems	Predominantly experimental validation; limited evidence of large-scale industrial deployment; no explicit assessment of blockchain overhead	Data sharing standardization framework showing reduced packet loss and improved compliance indicators

Authors (Year)	Title	Identified Limitations	Advances / Positive Results
Afrin et al. (2025)	Industrial Internet of Things: Implementations, Challenges, and Potential Solutions Across Various Industries	Persistent gaps in interoperability and scalability across IIoT domains (edge/fog/cloud, protocols, and integration)	Consolidated analysis of implementation challenges and solution directions across industrial sectors
Mahendran et al. (2025)	PRISM-IIoT: A Holistic Approach for Privacy Preservation in Industrial IoT Using Advanced Cryptography and Blockchain-enabled Auditability Framework	Interoperability issues with legacy industrial systems lacking support for crypto/blockchain layers	Blockchain-enabled auditability and accountability with integrity gains and latency aligned with industrial logging requirements
Kim and Vong (2025)	Secured Network Architectures Based on Blockchain Technologies: A Systematic Review	Limited laboratory evaluation of latency/throughput; field limitations include scalability, vulnerabilities, and compliance	Structured taxonomy of blockchain network architectures and interoperability-related perspectives across domains
Karankar and Seth (2025)	An IoT System for Access Control Using Blockchain and Message Queuing System	Performance bottlenecks, congestion in blockchain components, and scalability constraints for heterogeneous devices	Hybrid architecture with experimental results indicating improvements in execution time and load balancing

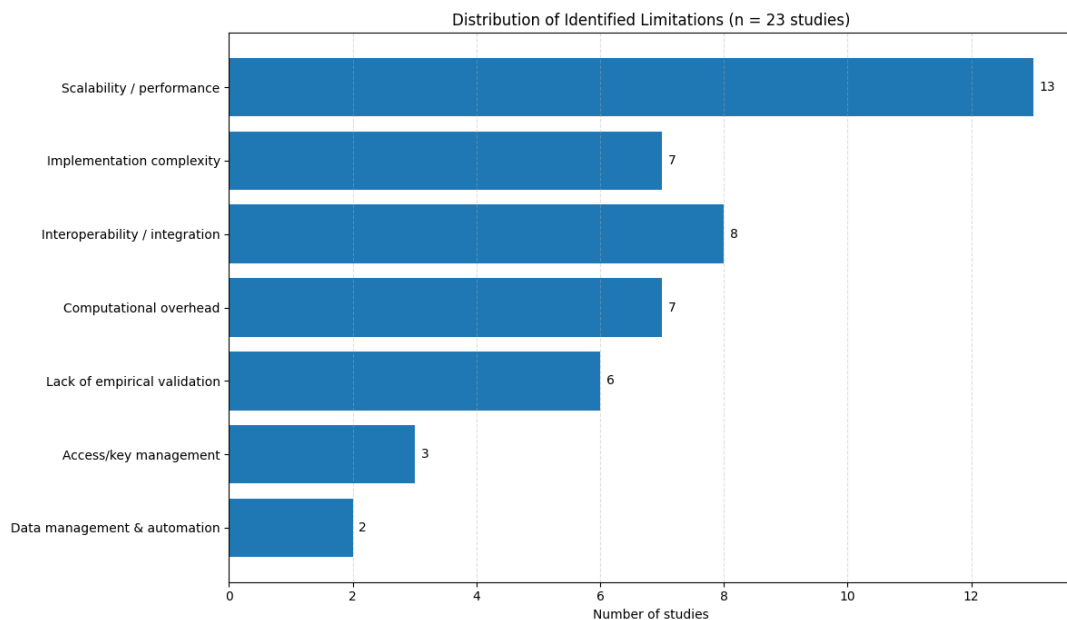
Source: Author's own work.

The analysis of the limitations identified in the studies reveals that, for the most part, the challenges are related to scalability, computational complexity, and the need for robust infrastructure. Additionally, the management of large volumes of decentralized data and the automation of contracts also stand out as areas that require further study. The absence of scalability

tests in real-world scenarios and delays in communication between components under high loads also prove to be frequent challenges.

To complement the qualitative synthesis presented in Table 4.4, the limitations reported in the 23 selected studies were grouped into analytical categories. Since some limitations are compound and refer to more than one issue simultaneously, a non-exclusive classification approach was adopted. Therefore, each study could be associated with one or more categories, preserving the meaning of the original findings while allowing the recurrence of each limitation type to be observed.

Figure 4.2: Distribution of identified limitations in the selected studies



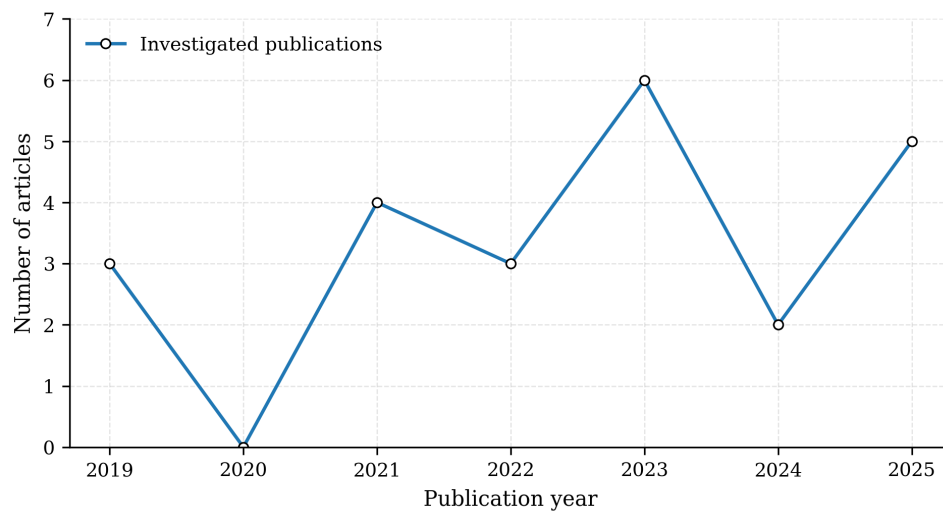
Source: Author's own work.

Figure 4.2 presents the distribution of the main limitations identified in the selected studies. The results indicate that scalability and performance issues are the most frequent limitation, appearing in 13 studies. Interoperability and integration challenges were also recurrent, appearing in 8 studies, followed by implementation complexity and computational overhead, each identified in 7 studies. Lack of empirical validation was observed in 6 studies, reinforcing that several proposals are still evaluated mainly in controlled or limited environments. Access and key management, as well as data management and automation, appeared less frequently, but remain relevant challenges in architectures involving blockchain, ABE, and IIoT data sharing.

This distribution reinforces that access management, interoperability, and data privacy are not isolated issues in the literature, but are closely related to broader architectural concerns.

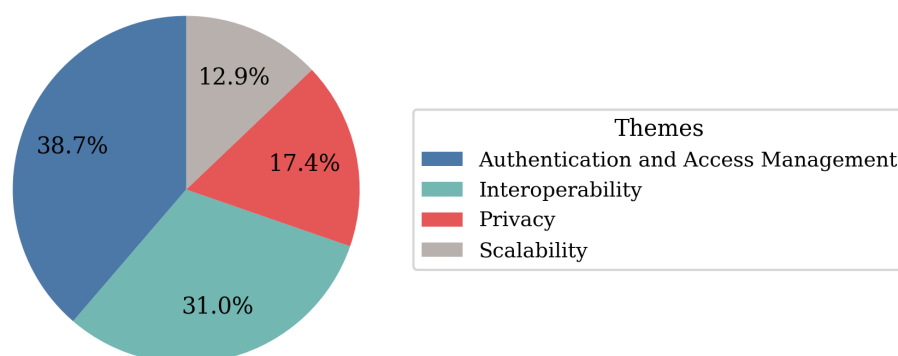
Although the searches considered, following the delimited exclusion criteria, only works published in the last six years, after applying the adopted protocol and using the methods of Kitchenham & Brereton (2013), it is noted that predominantly, quite recent works were analyzed. Observing this temporal analysis of the publications investigated, as well as the predominance of topics within the results obtained, is important to offer a comprehensive view of the evolution of research on the topic of this SLR. Next, in Figure 4.3 and Figure 4.4, two graphs are presented that illustrate, respectively, these analyses.

Figure 4.3: Temporal Evolution of Investigated Publications



Source: Author's Figure

Figure 4.4: Interpretation of results



Source: Author's Figure

The results presented point to a significant increase in the number of studies on blockchain in recent years, indicating the direction in which IIoT research is heading. It was also observed

that the most recurring themes include authentication and access management, interoperability, and privacy, representing the largest share of academic interest. In this classification, interoperability mainly refers to works that address communication architectures, distributed systems, and data exchange mechanisms. These findings reinforce the relevance of flexible access control and secure data exchange between organizations as recurring needs in IIoT environments.

4.7 Conclusion

The combination of blockchain with ABE presents a viable solution to address challenges related to interoperability and transaction privacy in IoT environments. By implementing smart contracts, most processes are automated, reducing the need for manual intervention and increasing trust and operational efficiency. Attribute-based encryption technology, ABE, ensures that only specific devices and users can access sensitive business information. This combination allows for granular access control and raises the level of information privacy, driving the creation of new business models involving IoT architecture.

Other research topics need to be explored to address the identified gaps. First, the scalability and performance of the proposed solution in a real industrial environment need to be carefully tested, as the operational conditions there hardly compare to controlled test environments. Investigating the integration of other emerging technologies, such as AI and machine learning in IIoT systems, can further increase the efficiency and security of the proposed architecture. They have the potential to offer proactive monitoring and anomaly detection. Considering the number of different devices involved, it is important to quantify the continuous computational expenditure involved in the process of creating and managing encryption keys for the ABE architecture employed in the proposed solution. Case studies in different industrial sectors can be useful to validate the applicability and effectiveness of the solution, as well as allow new business models to emerge.

Chapter 5

SYSTEM ARCHITECTURE

The proposed architecture adopts a hybrid on-chain/off-chain model, designed to ensure the integration of the technologies involved in the solution. Beyond interoperability, following the perspective adopted in this work of supporting data exchange between organizations with independent domains, the main objective is to preserve the ability of each participating organization to manage the privacy of shared data in a granular way.

The system integrates the following main components: IIoT devices acting as publishers and consumers, an ABE Cryptographic Module implemented with OpenABE, an MQTT broker for off-chain communication, a Python-based orchestrator acting as an Attribute Authority (AA), a Hyperledger Besu blockchain node, and a smart contract responsible for policy registration and key governance.

Considering the computational and energy costs discussed in Section 2.2, the proposed architecture does not assume that all IIoT sensor nodes directly perform ABE operations. ABE encryption, decryption, and key management procedures require a minimum execution environment with sufficient processing capacity, memory availability, and energy resources.

In this framework, these operations are assigned to an ABE Cryptographic Module, defined as a logical component responsible for applying cryptographic protection to IIoT payloads before their publication to the MQTT broker. This module is deployed within the trusted domain of the data-producing organization and may be implemented as part of the orchestration layer, a containerized service, or another computational component with adequate resources. In the current prototype, this role is performed by the Python-based orchestration layer, which receives sensor measurements, invokes the OpenABE container, and handles the protected payload generated by the cryptographic process.

Thus, battery-powered or resource-constrained sensors primarily act as data producers, while ABE operations are performed by a framework component with adequate computational resources. This design preserves the use of ABE for data-centric access control without

requiring constrained IIoT devices to execute expensive cryptographic operations directly.

In this model, sensitive data is not stored on the blockchain. Instead, the blockchain layer maintains immutable records of device registrations, access policies, and cryptographic key grant requests. Confidentiality is ensured off the blockchain through ABE, while authorization governance and auditability are ensured on the blockchain through the execution of smart contracts.

5.1 Use Case



To illustrate the problem addressed in this dissertation and the proposed architecture, consider a healthcare monitoring scenario in which a hospital collects patient data via wearable sensors. The hospital acts as the data owner and is, therefore, responsible for defining which entities may access specific information.

Within the hospital itself, professionals may require access to clinical measurements, provided their attributes (e.g., Role: Doctor and Department: Cardiology) satisfy the access policy. Furthermore, organizations external to the hospital may require access to different datasets. A health insurance provider might need only metadata related to device usage, or billing to validate and determine service coverage. The equipment manufacturer might require information regarding the device's status, such as battery logs, sensor quality, or calibration data, without accessing the patient's identity or clinical measurements.

In this scenario, the same sensing infrastructure can generate information relevant to multiple organizations, yet each organization must access only the data consistent with its role and purpose. Therefore, access control must be granular, auditable, and independent of a single trusted intermediary.

Although this use case is presented within the healthcare domain, the implemented prototype validates the same architectural principles in a controlled IIoT environment, utilizing MQTT, Hyperledger Besu, OpenABE, ECIES, and sensor payloads collected during the experimental evaluation.

5.2 Proposed Architecture Design

The operational design of the architecture is illustrated in Figure 5.1. The system is structured in two logically separate layers: an off-chain operational layer and an on-chain governance

layer.

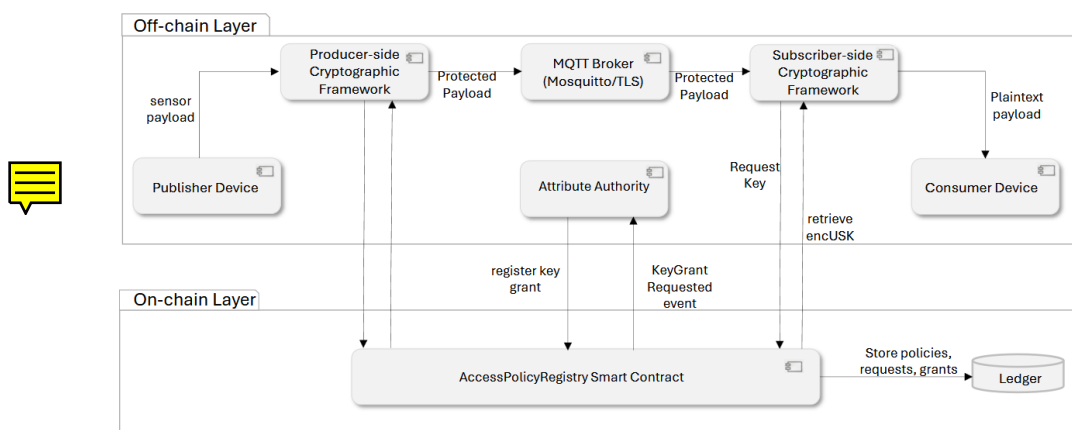
The off-chain layer contains the publish and consume devices, the MQTT broker, the Attribute Authority, and the cryptographic frameworks. In this layer, the publish device sends the sensor payload to the Cryptographic Framework on the Producer side. This framework, in turn, is responsible for encrypting the data according to the associated access policy and publishing the resulting protected payload to the MQTT broker.

The MQTT broker operates exclusively as a publish/subscribe routing mechanism. It receives the protected payload and delivers it to the Cryptographic Framework on the Subscriber side. Therefore, the broker is not responsible for performing authorization checks or enforcing access decisions.

The on-chain layer is implemented using Hyperledger Besu, where a smart contract named *AccessPolicyRegistry* was developed. This contract registers organizational attributes and access policies, as well as access requests and encrypted key grants. The blockchain does not perform ABE cryptographic operations. Instead, it coordinates access-related records and stores governance metadata in an auditable manner.

This organization separates cryptographic communication, governance, and enforcement responsibilities. This separation also prevents the publisher and consumer devices from having to perform key cryptographic procedures directly.

Figure 5.1: Logical Component Diagram of the Proposed Architecture



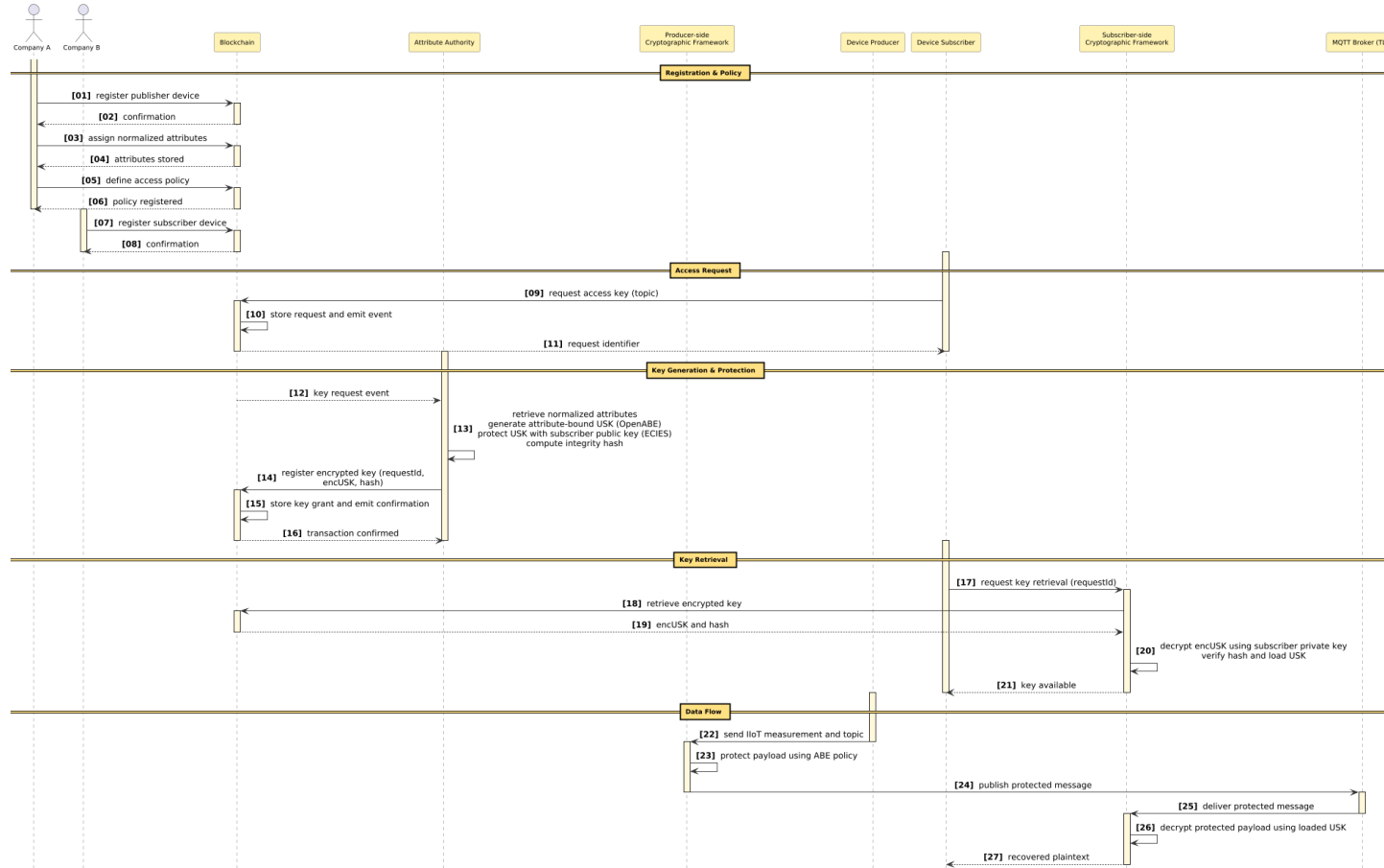
Source: Author's Figure

5.2.1 Sequence Diagrams

The operational sequence of the proposed architecture is illustrated in Figure 5.2. In light of recent studies and after analysis of the technologies involved, the architecture was systematically organized into five main phases. By decoupling identity registration and policy definition from the data transmission itself, the system allows that complex cryptographic operations remain outside the blockchain, while maintaining a verifiable audit log on the ledger and mitigating bottlenecks related to computational limitations.



Figure 5.2: Proposed Architecture Design



Source: Author's Figure

In the *Registration and Policy* phase, the participating organizations register producer and subscriber devices on the blockchain. Company A registers the producer device, assigns the normalized attributes associated with the device or organizational entity, and defines the access policy related to a specific MQTT topic. Company B registers the subscriber device that may later request access to protected data. At this stage, the blockchain acts as an auditable registry for identities, attributes, access policies, and governance events.

In the *Access Request* phase, the subscriber device requests access to a specific MQTT topic. This request does not directly provide access to the protected data. Instead, it creates an on-chain request record and emits an event that can be observed by the Attribute Authority. Therefore, the blockchain coordinates the key provisioning workflow and provides traceability, but does not execute the cryptographic operations.

In the *Key Generation and Protection* phase, the Attribute Authority observes the on-chain event and performs the cryptographic key provisioning process off-chain. At this point, the Attribute Authority retrieves the attributes already registered in the previous phase and generates an attribute-bound User Secret Key (USK) using OpenABE. The generated USK is then protected using ECIES with the subscriber's public key, and an integrity hash is computed. The encrypted key and its associated metadata are registered on the blockchain, creating an auditable record of the key grant.

In the *Key Retrieval* phase, the subscriber device requests the local Subscriber-side Cryptographic Framework to retrieve the encrypted key associated with the previous access request. The framework retrieves the encUSK and its hash from the blockchain, decrypts the encUSK using the subscriber's private key, verifies the integrity hash, and loads the recovered USK internally. In this design, the subscriber device does not manipulate the USK directly. It only receives the indication that the key material is available for subsequent decryption operations. This separation reduces the exposure of sensitive cryptographic material and keeps key handling inside the cryptographic framework.

Finally, in the *Data Flow* phase, the producer device sends the IIoT measurement and the corresponding topic information to the Producer-side Cryptographic Framework. This framework is responsible for protecting the payload according to the access policy associated with the MQTT topic and for publishing the resulting protected message through the MQTT broker over TLS. The MQTT broker only transports the protected message and does not enforce the cryptographic access decision. On the receiving side, the Subscriber-side Cryptographic Framework receives the protected message, uses the previously loaded USK to attempt decryption, and returns the recovered plaintext to the subscriber device only if the attributes embedded in

the USK satisfy the policy associated with the ciphertext.

The explicit inclusion of the Producer-side and Subscriber-side Cryptographic Frameworks represents an architectural decision adopted after evaluating where the cryptographic operations should occur. One possible alternative would be to execute ABE operations directly on the IIoT devices. However, this approach may be unsuitable for resource-constrained devices, especially when the number of attributes increases or when messages are produced frequently. Previous studies show that ABE can introduce relevant computational, memory, data-size, and energy costs in IoT and mobile environments (WANG et al., 2014; PERAZZO et al., 2021). Another alternative would be to delegate access control to the MQTT broker, but this would concentrate trust in an intermediary and would not provide data-centric confidentiality if the broker were compromised. Executing the cryptographic logic directly on the blockchain was also not adopted, since ABE operations are computationally expensive and should remain off-chain.

For these reasons, the proposed architecture assigns cryptographic processing to dedicated frameworks located within the trusted domains of the publishing and consuming organizations. The producer-side framework protects the payload before publication, while the subscriber-side framework manages the protected USK and performs decryption when the access policy is satisfied. This design preserves the role of IIoT devices as data producers or consumers, while the cryptographic frameworks handle the operations that require greater computational capability and sensitive key management.

The internal protection process follows the hybrid behavior commonly adopted by ABE implementations. Conceptually, the payload protection involves symmetric encryption of the data and ABE-based protection associated with the access policy. In practice, this process is abstracted by the OpenABE library. From a developer's perspective, and following the official OpenABE documentation, it is not necessary to manually implement the generation or protection of the AES key. Implementation involves calling the methods and manipulating the payload, access policy, and set of attributes required for encryption and decryption operations. It is worth noting that although attribute revocation is not implemented in the current prototype, the sequence represented in Figure 5.2 can be extended to support revocation strategies. In such an extension, the Attribute Authority could verify the revocation status, validity period, or policy version of each attribute before generating a new USK. Therefore, revocation would not be performed directly by the MQTT broker or by the blockchain execution layer, but by the off-chain Attribute Authority using the blockchain as an auditable source of attribute and policy status. This direction is consistent with revocation and lifecycle management challenges discussed in CP-ABE and fog-enabled IoT scenarios (PEÑUELAS-ANGULO et al., 2024).

5.3 Key Technologies

- MQTT Broker: Offers connectivity and data processing for IIoT data segments.
- Blockchain: Provides decentralization, transparency, and immutability to the solution.
- Attribute-Based Encryption (ABE): Ensures granular access and privacy management.
- Smart Contracts: Manage business rules, device registration, and cryptographic key distribution.

5.4 Experimental Environment

The experiments were conducted in a controlled local environment designed to support reproducibility and to isolate the main variables evaluated in this work. The infrastructure was executed on Ubuntu 24.04 LTS, running on a 12th generation Intel Core i7-1255U processor, with 10 cores, 12 threads, 16 GB of DDR4 RAM, and a 476 GB NVMe SSD.

The system prototype was organized using containerized components. The communication layer was implemented with an Eclipse Mosquitto MQTT broker, responsible for supporting the publish/subscribe communication model used by the IIoT devices and by the cryptographic modules. The blockchain layer was implemented with a Hyperledger Besu node configured as an EVM-compatible development network using `--network=dev`. The node exposed an HTTP JSON-RPC interface on port 8545 and operated with mining enabled to confirm transactions during the validation process. The integration logic was implemented in Python, using Paho-MQTT for MQTT communication and Web3.py for interaction with the blockchain layer.

The cryptographic layer was implemented using OpenABE, compiled from source code inside a custom Docker container. In the current evaluation, the OpenABE-based CP-ABE workflow was adapted into a compiled C++ module, referred to as `cpabe_split`, which separated the cryptographic process into four operations: setup, user key generation, encryption, and decryption. This separation allowed the experiments to represent more accurately the proposed architecture, in which the Attribute Authority issues User Secret Keys (USKs), the producer-side ABE Cryptographic Module encrypts the payload, and the subscriber-side ABE Module attempts to recover the protected data using the previously generated USK.

The setup operation generated the CP-ABE public and secret parameters. The key generation operation performed the role of the Attribute Authority by generating a USK associated with a given set of attributes. The encryption operation received a plaintext payload and an

access policy, producing an ABE ciphertext. The decryption operation attempted to recover the original payload using the generated USK. This structure allowed the experiments to evaluate not only whether the cryptographic process was functionally correct, but also how payload size, policy complexity, and message volume affect the size of the protected data and the processing time.

For the overhead evaluation, Python 3.12 scripts were used to invoke the OpenABE container through subprocess, collect execution times using `time.perf_counter()`, extract the generated ciphertext, encode it in Base64 when required, build the final ABE-protected payload, and store the resulting metrics in CSV files. Each CSV file was generated directly from real executions of the implemented prototype components.

In this evaluation, four data representations were considered. The *plaintext payload* corresponds to the original data before encryption. The *binary ABE ciphertext* corresponds to the binary output generated by OpenABE after encryption. The *Base64-encoded ABE ciphertext* corresponds to the textual representation of the binary ciphertext, required to include it in a JSON-based payload. Finally, the *ABE-protected payload* represents the final structure published after cryptographic protection, including the encoded ciphertext, the access policy, and auxiliary metadata required for transmission and recovery of the protected data.

Table 5.1 summarizes these representations.

Table 5.1: Data representations considered in the overhead evaluation.

Representation	Description
Plaintext payload	Original payload before cryptographic protection, such as a sensor measurement or a controlled JSON message.
Binary ABE ciphertext	Binary output produced by OpenABE after applying CP-ABE encryption to the plaintext payload.
Base64-encoded ABE ciphertext	Textual representation of the binary ciphertext, used to include the encrypted content in JSON-based messages.
ABE-protected payload	Final protected structure containing the encoded ciphertext, the access policy, and auxiliary metadata used for transmission and recovery.

Three complementary experiments were conducted. The first experiment evaluated the impact of plaintext payload size on spatial overhead. The second experiment evaluated the impact of access policy complexity, measured by the number of attributes in the policy. The third experiment evaluated the accumulated overhead produced when real ESP32/DHT22 sensor me-

asurements were individually protected before publication.

In addition to cryptographic overhead experiments, a proof of concept was implemented in an integrated manner through the module called `<integrated-poc>`. This module includes the smart contract `<AccessPolicyRegistry>`, Python scripts for blockchain initialization and access requests, an off-chain Attribute Authority, ECIES utilities for cryptographic key privacy, sensor data collection scripts, subscriber-side key recovery logic, runtime evidence files, and a Streamlit interface for guided execution. This framework was used to validate the complete workflow and will be detailed in the following chapters.

Chapter 6

RESULTS

This chapter presents the results obtained from the experimental plan executed using the proposed architecture. The focus is on the feasibility analysis of the technologies, the objective measurement of cryptographic performance, access privacy, and the behavior of the solution as the necessary access policies become more complex. After presenting these results, the interpretations and implications will be discussed in the following chapter.

6.1 System Integration and Proof-of-Concept Validation

The system integration validation involved all components of the architecture, including sensor data collection, CP-ABE payload protection with OpenABE, ECIES-protected key delivery, an off-chain Attribute Authority, and a local Hyperledger Besu blockchain network. The integration was conducted with a focus on verifying that the components could operate as a coherent end-to-end workflow.

The implemented proof of concept was organized in the `integrated-poc` module. This module includes the smart contract, orchestration scripts in Python, the Attribute Authority service, key retrieval logic, and runtime evidence files. To make the deliverable of this research less dependent on manual command injection, automate testing, and create a reproducible architecture workflow, an interface based on Streamlit, a free and open-source Python library, was implemented. The integration was validated in a local environment using an ESP32/DHT22 sensor stream, an Eclipse Mosquitto MQTT broker, an OpenABE Docker container, Python scripts based on Web3.py and Paho-MQTT, and a Besu Hyperledger node exposed via a JSON-RPC endpoint.

The smart contract `AccessPolicyRegistry` was used as the on-chain coordination and governance layer. The contract provides functions to register producers and consumers, associate access policies with protected topics, register and retrieve subscriber attributes, create access

requests, register protected messages and consumption records, and store encrypted key grants with integrity hashes. It is also important to note that, in this way, the blockchain was not used to store sensitive sensor data in plain text, but to provide an auditable layer for metadata related to access and key provisioning events.

The on-chain state required for validation was initialized by the bootstrap script `<02_bootstrap_onchain_state.py>`. This script registers the producer device, the subscriber device, the subscriber attributes, and the access policy associated with the protected topic. In the default configuration, the protected topic was defined as `<sensors/sensor_001/protected>`. The validation included scenarios in which the subscriber attributes satisfied the access policy and scenarios in which they did not. In both cases, the cryptographic behavior remained consistent with the expected CP-ABE enforcement model: the payload was recovered only when the attributes embedded in the subscriber key satisfied the policy associated with the ciphertext. The constructed interface allows the attributes to be changed in the initial step, for validation of the cryptographic behavior.

The sensor data collection step was implemented by the script `<collect_mqtt_sensor_encrypt.py>`. In this step, an ESP32 device connected to a DHT22 sensor published temperature and humidity readings in JSON format to the MQTT topic `<sensors/sensor_001/plain>`. The script validated the received payload, normalized the message, and invoked the OpenABE environment to encrypt the payload according to the associated policy. This step validated the connection between the sensor's physical data acquisition layer and the cryptographic protection mechanism used by the architecture.

After registering the parties involved and generating the protected payload, the `<request_access.py>` script creates an auditable access request on the blockchain for the protected topic. This request does not grant direct access to the plaintext data. Instead, it registers a pending access request in the smart contract, making it available for subsequent processing by the Attribute Authority.

The off-chain Attribute Authority is implemented in `<attribute_authority_service.py>`. This component retrieves access requests from the smart contract, obtains the subscriber attributes registered on the blockchain, and generates an OpenABE User Secret Key (USK) associated with those attributes. A relevant design decision is that the Attribute Authority does not fulfill the role of making the final access decision through a conditional rule at the application level. In the adopted CP-ABE model, the Attribute Authority issues a key based on the attributes registered for the subscriber, as represented in the sequence diagram presented in Section 5.2.1, while the actual access decision occurs during decryption. If the attributes embedded

in the USK satisfy the policy embedded in the ciphertext, decryption is successful; otherwise, the protected payload cannot be recovered.

Following the planned architecture and to avoid exposing the OpenABE USK on the blockchain, the generated key was exported from the OpenABE container, encoded, packaged as a JSON artifact, and secured using ECIES, a public-key cryptography scheme used in the Ethereum ecosystem, with the subscriber's public key. The resulting encrypted user secret key, named `<encUSK>`, was registered in the `<AccessPolicyRegistry>` contract via the key grant workflow, along with an integrity hash. This design keeps the blockchain responsible for the traceability and availability of the encrypted grant, while preserving the confidentiality of the ABE user secret key.

The subscriber-side recovery process was implemented by the script `<retrieve_encrypted_usk.py>`. This script retrieves the encrypted key grant from the smart contract, decrypts `<encUSK>` locally using the subscriber's ECIES private key, calculates the hash of the recovered artifact, compares it to the hash recorded on the blockchain, and restores the OpenABE user's secret key as a binary file. After this step, the restored OpenABE key is used to attempt to decrypt the current ciphertext.

End-to-end validated execution confirmed that a subscriber whose attributes meet the policy `<attr1orattr2>` was able to recover the original MQTT sensor payload generated by the ESP32/DHT22 stream. In contrast, when the subscriber's attributes were set to `<attr3>`, the OpenABE decryption process failed as expected.

To support reproducibility, additional scripts and validation mechanisms were incorporated into the proof of concept. The script `<check_environment.py>` verifies the Besu RPC endpoint, chain ID, deployed contract address, expected contract functions, OpenABE container, subscriber ECIES keys, topic policy, and subscriber attributes. The script `<run_full_demo.sh>` executes the main validation steps sequentially and stores logs, evidence files, and a structured execution summary in a timestamped runtime directory. A Streamlit interface, implemented in `<integrated-poc/ui/streamlit_app.py>`, was also developed to support step-by-step execution, full demonstration execution, log visualization, and comparison between authorized and unauthorized scenarios.

The integration results demonstrate that the proposed architecture is viable in the evaluated local environment. Validation demonstrated the combined operation of MQTT-based IIoT data collection, CP-ABE payload protection, blockchain-based policy and attribute registration, ECIES-protected key delivery, and subscriber-side payload retrieval. These results complement the cryptographic performance and overhead assessments presented in the following sections.

6.2 Cryptographic Performance

The architecture was evaluated focusing on identifying the computational overhead introduced by Attribute-Based Encryption during cryptographic operations. Experimental tests were conducted using three payload sizes (1 KB, 100 KB, and 1000 KB) and five policy configurations composed of 1, 10, 20, 30, and 40 distinct attributes.

The selection of payload sizes was defined to represent different IIoT communication scenarios. The 1 KB payload simulates light sensor readings, typically transmitted at high frequency (e.g., temperature, pressure, vibration). The 100 KB payload represents aggregated telemetry data or structured industrial messages. The 1000 KB payload simulates heavier industrial data transfers, such as batch logs, diagnostic reports, or enriched sensor streams.

The evaluation of different payload sizes considered experimental evidence in the literature. Domingues et al. (2024) demonstrate that payload sizing can impact MQTT transmission time and recovery behavior after network disconnections, showing measurable performance differences as message size increases. Along these lines, Gavriilidis et al. (2025) empirically evaluate TLS-enhanced MQTT under different message sizes, including payloads on the order of hundreds of kilobytes and megabytes, and report observable impacts on latency and resource utilization. These results highlight the need to consider potential performance influences from fluctuating payload sizes and, therefore, to evaluate cryptographic overhead under different payload magnitudes, rather than assuming a single fixed message size. Consequently, the adopted range of 1 KB, 100 KB, and 1000 KB allows for the evaluation of the proposed architecture under experimentally grounded IIoT communication conditions.

Similarly, the number of attributes was defined to represent increasing levels of complexity in the access policy. In industrial environments where the companies involved do not have a premise of mutual trust, the ways of managing the privacy of shared data can fluctuate greatly, depending on multiple contextual conditions, such as department, function, operational unit, or employee access level, for example. A single attribute represents a minimum access control condition. Configurations with 10, 20, 30, and 40 attributes simulate progressively more complex and granular access policies, allowing the evaluation of the cryptographic mechanism's behavior under increasing structural complexity of the access rule.

To mitigate transient effects of the system and increase the statistical stability of the results, for each combination of message size and number of attributes, 30 executions were performed, totaling 450 cryptographic measurements (3 sizes $\times$ 5 attribute configurations $\times$ 30 repetitions).

For each configuration, the average encryption time was calculated as shown in Equa-

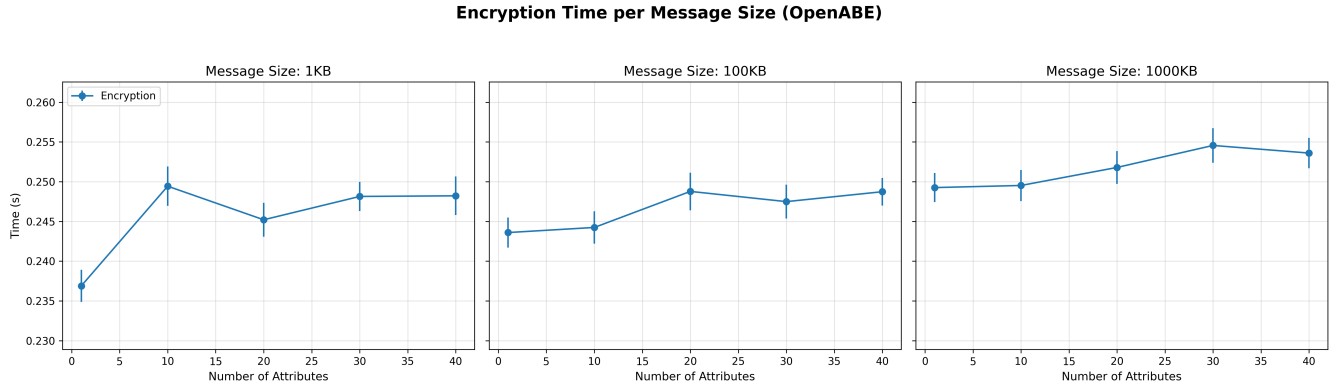
tion 6.1:

$$T = \frac{1}{n} \sum_{i=1}^n t_i \quad (6.1)$$

where $n = 30$ represents the number of executions and t_i is the encryption time observed in each run.

Figure 6.1 presents the average encryption time as a function of message size and policy complexity. The results show that encryption time increases moderately with policy complexity within the evaluated range, while variations in payload size produce comparatively smaller differences.

Figure 6.1: Encryption time by message size and number of attributes.



Source: Author's Figure

To better quantify the impact of the complexity of the attribute policy, Table 6.1 shows the percentage variation in the encryption time relative to the baseline configuration (1 attribute) for each payload size. The percentage variation was calculated as shown in Equation 6.2:

$$\% \Delta = \frac{T_{\text{current}} - T_{\text{baseline}}}{T_{\text{baseline}}} \times 100 \quad (6.2)$$

where T_{baseline} corresponds to the mean encryption time using a single attribute.

Table 6.1: Percentage variation in encryption time relative to the baseline configuration (1 attribute).

Message Size	10 attrs	20 attrs	30 attrs	40 attrs
1 KB	+5.36%	+3.65%	+4.80%	+4.87%
100 KB	+0.27%	+2.22%	+1.55%	+2.09%
1000 KB	+0.03%	+0.92%	+2.08%	+1.81%

Source: Author's own work.

The results indicate that increasing the policy complexity from 1 to 40 attributes leads to a maximum observed increase of approximately 5% for 1 KB payloads and about 2% for larger payloads. This demonstrates that, although encryption time increases with policy complexity, the magnitude of the increase remains limited within the evaluated range. Furthermore, the relatively small percentage differences between message sizes indicate that, in terms of encryption time, the evaluated cryptographic cost is more influenced by policy structure than by payload volume.

Although large-scale deployments involving thousands of devices were not evaluated at this stage, a consistent and bounded growth pattern in relation to the number of attributes suggests predictable scalability of the cryptographic component.

6.3 Cryptographic Overhead Evaluation

The cryptographic overhead evaluation focused on measuring the increase in data volume introduced by ABE-based protection and the corresponding processing time required for encryption and decryption. In this work, spatial overhead is defined as the increase in data size produced when the original plaintext payload is transformed into an ABE-protected payload.

The spatial overhead percentage was calculated as shown in Equation 6.3:

$$\text{Overhead}(\%) = \frac{S_{\text{protected}} - S_{\text{plain}}}{S_{\text{plain}}} \times 100 \quad (6.3)$$

where S_{plain} represents the size, in bytes, of the original plaintext payload, and $S_{\text{protected}}$ represents the size, in bytes, of the ABE-protected payload.

The overhead ratio was calculated as shown in Equation 6.4:

$$\text{Ratio} = \frac{S_{\text{protected}}}{S_{\text{plain}}} \quad (6.4)$$

This metric indicates how many times larger the protected payload is when compared to the original plaintext payload. It is important to highlight that the measured processing times correspond to the end-to-end invocation time of the cryptographic module.

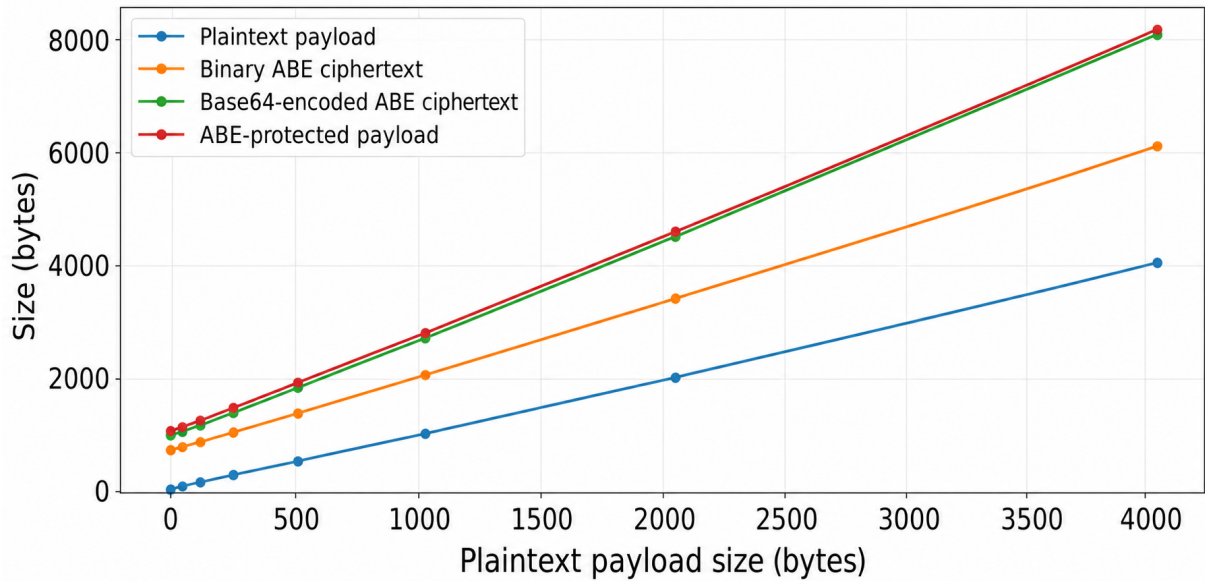
6.3.1 Spatial Overhead by Payload Size

The first experiment evaluated how the size of the original payload affects the spatial overhead introduced by ABE-based protection. The access policy was kept fixed as `attr1` or `attr2`, and the USK was generated with the attribute set `|attr1`. Since the attribute `attr1` satisfies the defined policy, all decryption operations were expected to succeed.

The evaluated payloads included one real sensor baseline payload and eight controlled JSON payloads. The real sensor baseline corresponded to a measurement generated by the ESP32/DHT22 device, with an average plaintext size of 24 bytes. The controlled payloads were generated with sizes of 32, 64, 128, 256, 512, 1024, 2048, and 4096 bytes. For each payload size, 10 executions were performed, resulting in 90 encryption and 90 decryption operations.

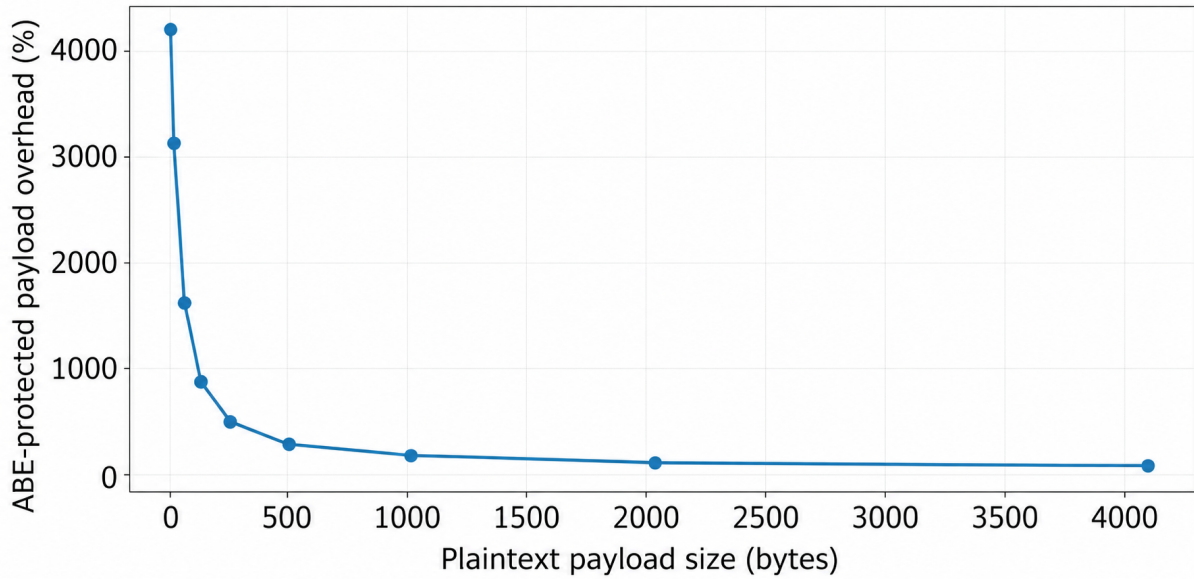
For each execution, the following metrics were recorded: plaintext payload size, binary ABE ciphertext size, Base64-encoded ABE ciphertext size, ABE-protected payload size, overhead ratio, overhead percentage, encryption time, decryption time, and decryption success status.

Figure 6.2 shows the growth of the data size across the evaluated representations.

Figure 6.2: Data size growth by payload size.**Source: Author's Figure**

The results show that the ABE-protected payload is larger than the original plaintext in all evaluated cases. This effect is proportionally more significant for small payloads, since the cryptographic structures, Base64 encoding, access policy, and auxiliary metadata introduce an overhead that becomes large when compared with small sensor measurements. For the real sensor baseline, the plaintext payload had 24 bytes, while the corresponding ABE-protected payload reached 1049 bytes. This represents a significant increase in data volume when individual sensor measurements are protected separately.

Figure 6.3 presents the overhead percentage as a function of the plaintext payload size.

Figure 6.3: ABE-protected payload overhead by payload size.

Source: Author's Figure

The overhead percentage is higher for small payloads because the cryptographic protection introduces structures that are required independently of the original message size. These structures include the ABE ciphertext, policy-related components, Base64 encoding, and auxiliary fields included in the protected payload. As a result, when the original payload is small, this fixed protection structure becomes proportionally much larger than the plaintext itself.

For the real sensor payload, the overhead reached 4270.83%. For a 4096-byte payload, the overhead decreased to 102.47%. This reduction does not indicate that the protected payload becomes smaller in absolute terms. Rather, it shows that the fixed part of the cryptographic and encoding overhead is diluted as the plaintext payload size increases.

Therefore, the first experiment indicates that applying ABE to very small IIoT measurements can introduce substantial spatial overhead. This result is particularly relevant in scenarios where sensors publish small messages at high frequency.

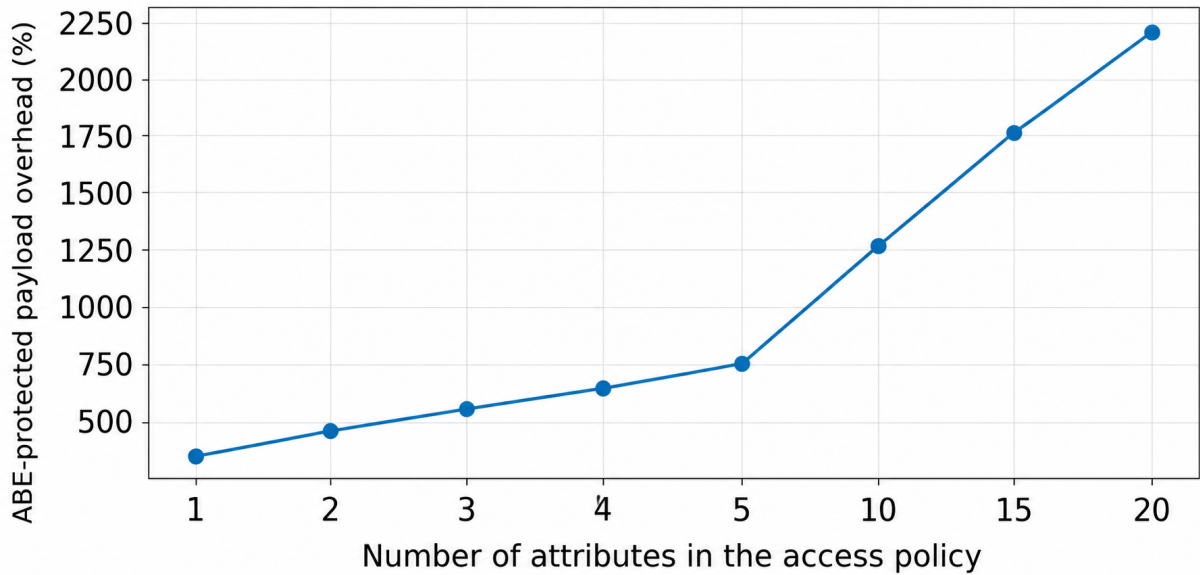
6.3.2 Impact of Access Policy Complexity

The second experiment evaluated the impact of access policy complexity on the size of the protected data and on the cryptographic processing time. In this experiment, the plaintext payload size was fixed at 256 bytes, while the number of attributes in the access policy was varied.

The evaluated policies contained 1, 2, 3, 4, 5, 10, 15, and 20 attributes. The single-attribute configuration was used as the baseline. Policies with two or more attributes were composed using the AND logical operator, requiring all attributes in the policy to be satisfied. For example, the policy with three attributes was represented as attr1 and attr2 and attr3. For each policy configuration, a compatible USK was generated with the corresponding set of attributes, ensuring that decryption was expected to succeed. For each configuration, 10 executions were performed, resulting in 80 encryption and 80 decryption operations.

Figure 6.4 presents the overhead percentage of the ABE-protected payload as the number of attributes in the access policy increases.

Figure 6.4: ABE-protected payload overhead by access policy complexity.



Source: Author's Figure

The results show that policy complexity directly affects the spatial overhead of the protected data. With a fixed plaintext payload of 256 bytes, the ABE-protected payload increased as more attributes were included in the access policy. With one attribute, the protected payload had an average size of 1216 bytes, corresponding to an overhead of 375.00%. With 20 attributes, the protected payload reached an average size of 6069 bytes, corresponding to an overhead of 2270.70%.

This behavior is consistent with CP-ABE schemes, since the ciphertext structure includes components associated with the access policy. As the policy becomes more complex, more cryptographic material must be included in the ciphertext, increasing the final size of the pro-

tected payload.

The experiment also shows that fine-grained access control introduces a trade-off. More expressive policies allow more precise control over which entities can decrypt the data, but they also increase the amount of information that must be transmitted and processed. Therefore, in multi-organizational IIoT scenarios, policy design should consider not only access control requirements, but also the communication and storage overhead introduced by highly granular policies.

6.3.3 Accumulated Overhead with Real Sensor Measurements

The third experiment evaluated the accumulated overhead produced when real IIoT measurements are protected individually. Unlike the previous experiments, this evaluation used real payloads generated by an ESP32 device connected to a DHT22 temperature and humidity sensor. The device published measurements to the MQTT topic `sala/sensor`. Each received measurement was normalized, encrypted by the producer-side ABE Cryptographic Module, and then decrypted by the subscriber-side ABE Module.

The access policy was kept fixed as `attr1` or `attr2`, and the USK was generated with the attribute set `|attr1`. Therefore, all decryption operations were expected to succeed. The experiment evaluated three message volumes: 30, 60, and 120 real sensor messages, totaling 210 processed messages.

The overhead percentage was calculated by comparing the accumulated size of the ABE-protected payloads with the accumulated size of the corresponding plaintext payloads, as shown in Equation 6.5:

$$\text{Overhead}(\%) = \frac{S_{\text{protected}} - S_{\text{plain}}}{S_{\text{plain}}} \times 100 \quad (6.5)$$

where S_{plain} represents the accumulated size of the plaintext payloads, and $S_{\text{protected}}$ represents the accumulated size of the ABE-protected payloads for each evaluated message volume.

Table 6.2 summarizes the accumulated results.

Table 6.2: Accumulated overhead for real sensor measurements.

Messages	Plaintext bytes	Protected bytes	Avg. plain bytes/msg	Avg. protected bytes/msg	Overhead	Success
30	720	31470	24	1049	4270.83%	30/30
60	1440	62940	24	1049	4270.83%	60/60
120	2880	125880	24	1049	4270.83%	120/120

Source: Author's own work.

The results show that all 210 real sensor messages were successfully encrypted and decrypted, with no invalid messages and no decryption failures. This confirms that the subscriber-side ABE Module was able to recover the original measurements when the USK satisfied the access policy.

The overhead percentage remained constant across the evaluated message volumes because the average plaintext size and the average ABE-protected payload size remained the same for each individual measurement. Each real sensor measurement had an average plaintext size of 24 bytes, while the corresponding ABE-protected payload had an average size of 1049 bytes. Therefore, increasing the number of messages increased both the accumulated plaintext size and the accumulated protected size proportionally.

However, the results also show that the total protected data volume accumulates linearly when each measurement is protected individually. This occurs because each sensor reading generates an independent ABE-protected payload, including its own ciphertext, encoding, policy information, and auxiliary metadata. Therefore, doubling the number of protected measurements approximately doubles the total protected data volume.

This behavior confirms that applying ABE to individual small measurements can significantly increase the total amount of data transmitted or stored, even when the relative overhead percentage remains constant across message volumes. From an architectural perspective, this result does not invalidate the proposed approach. Rather, it indicates that the granularity of cryptographic protection must be carefully evaluated. Protecting each individual measurement provides fine-grained confidentiality and access control, but it may be inefficient in scenarios with high-frequency sensing and very small payloads. In practical IIoT deployments, alternative strategies such as protecting batches of measurements, time-windowed payloads, or aggregated messages may reduce the relative overhead while preserving data-centric access control.

6.3.4 Processing Time Analysis

In addition to spatial overhead, the experiments also recorded the time required to invoke the cryptographic operations. Since the implementation executes OpenABE inside a container and invokes the cryptographic module through Python scripts, the measured values represent end-to-end cryptographic module invocation time. Therefore, these values include not only the cryptographic operation itself, but also the execution overhead associated with the implemented prototype.

In the payload-size evaluation, the average encryption time remained approximately between 65 ms and 74 ms, while the average decryption time remained approximately between 76 ms and 81 ms across the evaluated payload sizes. No strong linear increase was observed as the plaintext payload size increased up to 4096 bytes. This suggests that, in the evaluated implementation, a relevant part of the processing time is associated with the invocation of the containerized module and the fixed cost of the cryptographic operation, rather than only with the payload size.

In the policy-complexity evaluation, the processing time increased as the number of attributes in the access policy grew. Figure 6.5 presents the mean encryption and decryption times observed for each access policy configuration.

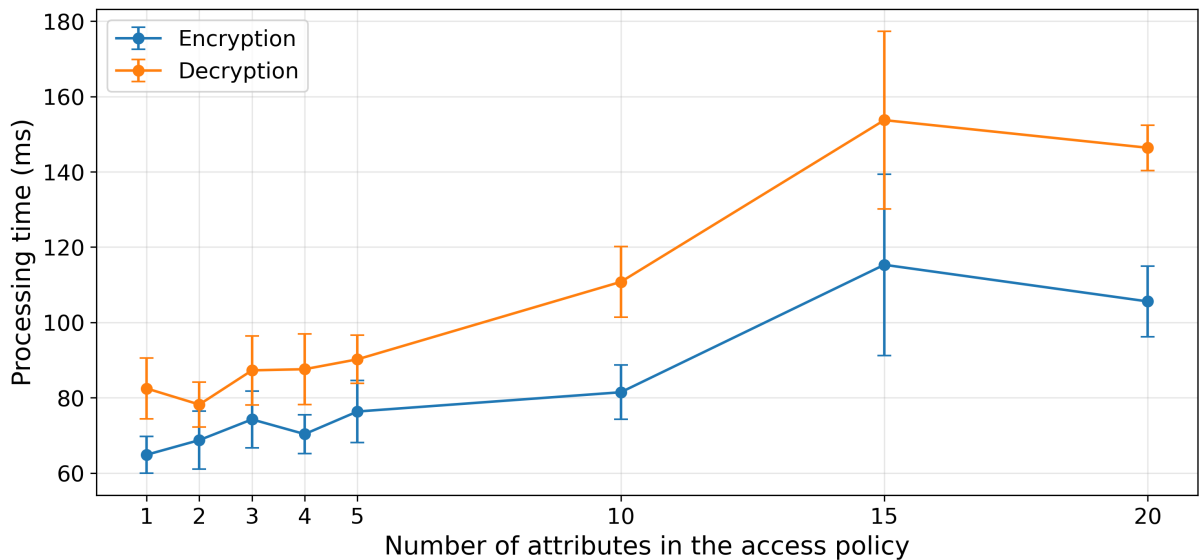


Figure 6.5: Processing time by access policy complexity.

As shown in Figure 6.5, decryption time is generally higher than encryption time and becomes more affected as policy complexity increases. Encryption time varied from approximately 64.86 ms with one attribute to 105.58 ms with 20 attributes, while decryption time varied from

approximately 82.47 ms to 146.38 ms. This behavior is consistent with the additional verification required to determine whether the attributes embedded in the USK satisfy the access policy associated with the ciphertext.

In the real-sensor message evaluation, the average processing time remained stable across the evaluated message volumes. The average encryption time remained between approximately 74 ms and 77 ms per message, while the average decryption time remained between approximately 79 ms and 80 ms per message. The accumulated processing time increased proportionally to the number of processed messages, as expected.

These results indicate that the proposed architecture is functionally viable in the evaluated local environment. However, they also show that invoking ABE-based protection for each individual small measurement introduces a measurable processing cost. Therefore, the use of ABE in IIoT scenarios should be aligned with the expected publication frequency, payload size, policy complexity, and latency requirements of the application.

6.4 Privacy and Access Control

The data published during the evaluated scenarios was correctly encrypted before publication, ensuring that only consumer devices possessing the set of attributes compatible with the defined access policies could decrypt the messages. Furthermore, decryption tests were performed using keys that did not meet the access policies, and all failed, confirming that decryption succeeds only when the attributes associated with the USK satisfy the access policy embedded in the ciphertext. This behavior was also consistent across different payload sizes and policy complexities. By implementing access control at the cryptographic layer, the architecture ensures that data confidentiality is preserved regardless of the communication infrastructure. The MQTT broker only handled encrypted payloads, with no exposure of plaintext data in intermediate components.

6.5 Scalability and Data Volume

Scalability in the proposed architecture was analyzed from two complementary perspectives: (i) cryptographic scalability, associated with the structural complexity of ABE access policies, and (ii) data volume scalability, related to the size of encrypted messages transmitted through the IIoT communication layer.

6.5.1 Cryptographic Scalability

Cryptographic scalability was evaluated by progressively increasing the number of attributes in the access policy from 1 to 40 attributes, while maintaining controlled execution conditions (30 runs per configuration). As demonstrated in Section 6.2, the increase in encryption time remained bounded within approximately 5% for small payloads (1 KB) and below 2.1% for larger payloads (100 KB and 1000 KB).

These results indicate that during the data encryption stage, the computational overhead increases mainly as a function of the complexity of the applied attribute policy, but within a predictable range. The growth trend observed in all configurations suggests that the cost of encryption scales proportionally to the structural expansion of the access rule, without exhibiting unstable behavior within the evaluated range.

6.5.2 Data Volume Scalability

Data volume scalability was analyzed by considering both the processing behavior observed for different payload sizes and the spatial overhead introduced by ABE-based protection. The results presented in Section 6.3 show that the ABE-protected payload increases in absolute size as the plaintext payload grows. However, the relative overhead is higher for small payloads, since the fixed cryptographic structure, encoding, policy information, and auxiliary metadata represent a larger proportion of the original data.

The results also show that when individual sensor measurements are protected separately, the accumulated protected data volume grows proportionally to the number of messages. This behavior indicates that data volume scalability depends not only on payload size, but also on the granularity at which cryptographic protection is applied.

Therefore, while the processing-time results indicate stable behavior within the evaluated configurations, the spatial overhead results show that high-frequency transmission of small protected messages may increase communication and storage costs. This reinforces the need to evaluate batching, time-windowed payloads, or aggregated messages in future deployments.

It is important to note that scalability in terms of simultaneous device concurrency or high-throughput industrial deployments was not evaluated in this phase. However, the controlled experimental results provide evidence about the behavior of the architecture in relation to policy complexity, payload size, and accumulated protected data volume.

Chapter 7

DISCUSSION

In the context of the results obtained, this chapter will discuss the implications related to them. The focus of the discussions will be to examine the characteristics of the performance obtained, analyzing the trade-offs of the architecture and identifying limitations and open challenges related to the proposed solution.

7.1 Interpretation of Results

The limited and predictable increase in encryption and decryption time as policy complexity increased indicates that ABE can be integrated into the proposed architecture without introducing unstable behavior in the evaluated conditions. It is also important to distinguish processing time from spatial overhead. In terms of processing time, payload size had a smaller influence when compared to policy complexity. However, in terms of data volume, small payloads were proportionally more affected by the cryptographic protection.

The overhead evaluation showed that the ABE-protected payload is significantly larger than the original plaintext payload, especially for small IIoT messages. This behavior occurs because ABE protection introduces cryptographic structures, policy-related components, encoding, and auxiliary metadata that are required independently of the original payload size. Therefore, when the original sensor measurement is small, this fixed structure becomes proportionally larger than the plaintext itself.

This result is relevant for IIoT scenarios in which sensors publish small measurements at high frequency. In the experiment using real ESP32/DHT22 measurements, each plaintext payload had an average size of 24 bytes, while the corresponding ABE-protected payload had an average size of 1049 bytes. Although all messages were correctly encrypted and decrypted, the accumulated data volume increased linearly as individual measurements were protected separately. This indicates that protecting each measurement individually provides fine-grained

access control, but may increase the volume of data transmitted, processed, or stored.

The impact of access policy complexity was also observed in the spatial overhead results. With a fixed plaintext payload, increasing the number of attributes in the access policy increased the size of the ABE-protected payload. This behavior is consistent with the structure of CP-ABE schemes, in which the ciphertext includes components associated with the access policy.

From an architectural perspective, maintaining access control in a cryptographic layer, rather than the broker or application layer, strengthens security guarantees in environments involving multiple organizations without an implicit trust relationship, that is, access policies are imposed directly on encrypted data, regardless of the transport medium or the intermediary entity responsible for communication. In this way, even if the broker or consuming application is compromised or operates outside a shared trust domain, access to the content remains conditioned exclusively on possession of the cryptographic attributes required by the defined policy.

The results also show that the proposed architecture involves a trade-off between access control granularity and operational cost. ABE provides data-centric and fine-grained access control, but its application to each individual small measurement can generate significant spatial overhead. Thus, the decision to protect data at the level of individual measurements, batches, time windows, or aggregated messages should be aligned with the privacy requirements, publication frequency, payload size, and latency constraints of the IIoT scenario.

7.2 Advantages of the Proposed Solution

The proposed solution offers privacy and security through the use of ABE, which ensures that only entities whose attributes satisfy the defined access policies can access specific shared data. In addition, the architecture supports controlled data exchange between organizations, even when there is no prior trust relationship between the parties involved, integrating MQTT-based communication, blockchain-based auditability, and access control with ABE.

An advantage is that the architecture does not require constrained IIoT sensors to execute ABE operations directly. In the evaluated prototype, cryptographic operations are assigned to the ABE Cryptographic Module, which is executed in an environment with adequate computational resources. This design allows the use of ABE for data-centric protection while avoiding the direct execution of expensive cryptographic operations on low-capacity sensing devices.

7.3 Limitations and Challenges

Despite the positive results, some limitations and challenges remain. First, the experiments were conducted in a local environment using general-purpose computing hardware. The impact of ABE on different industrial execution environments, such as embedded controllers, industrial gateways, and constrained edge devices, was not evaluated and represents an important direction for future testing.

Second, the experimental evaluation focused on cryptographic behavior, spatial overhead, and controlled message volume. Therefore, system-level scalability aspects, such as concurrent devices, multiple organizations, high-frequency message streams, and throughput under distributed deployment conditions, were outside the scope of this work.

Third, the overhead experiments showed that protecting small measurements individually can substantially increase the transmitted data volume. This indicates that the granularity of cryptographic protection must be defined according to the characteristics of the application. In scenarios with high-frequency sensing and small payloads, batching or aggregating multiple measurements before applying ABE may reduce the relative overhead.

Fourth, memory consumption was not directly measured in the current evaluation. The experiments recorded spatial overhead and cryptographic module invocation time, but did not capture peak memory usage during encryption and decryption. Therefore, memory profiling remains a limitation of the current stage and should be addressed in future evaluations using process-level measurement tools capable of capturing the maximum resident set size of the cryptographic operations.

Fifth, the current architecture adopts an Attribute Authority responsible for issuing attribute-bound User Secret Keys. Although this authority does not operate arbitrarily, since the key provisioning workflow is triggered by on-chain events, it still represents a logically centralized component in the trust model. This is an open and relevant challenge in multi-organizational IIoT scenarios. Alternative approaches, such as Multi-Authority ABE and Self-Sovereign Identity, were analyzed during the Systematic Literature Review, but their implementation was not addressed in the current prototype.

In addition to the limitations observed in the experimental evaluation, the architecture also presents open challenges related to the management of the attribute lifecycle and access rights, as described in Subsection 7.3.1.

7.3.1 Attribute Revocation

Another important challenge relates to attribute lifecycle management, specifically attribute revocation. In cryptographic ABE mechanisms, revocation is a classic impasse, as it is not limited to removing an attribute from a registry. Once a User Secret Key (USK) has been properly issued to a consumer, this key may remain stored locally and still satisfy previously defined access policies. For this reason, if a company leaves a collaborative process, if a business contract expires, if a device is compromised, or if an organizational role is changed, the system must prevent the affected entity from continuing to decrypt data. Hur & Noh (2011) discuss this problem in the context of third-party data systems based on CP-ABE and show in their work that revocation becomes more complex because in many scenarios attributes are commonly shared by distinct groups of users. In their analysis, the authors show that time-based revocation can create a vulnerability window: a user, even if revoked, can continue accessing data until the next key update, while a newly authorized user can access previous data unless the ciphertexts and keys are properly updated. The authors also associate this revocation approach with scalability costs, since the key update material may need to be distributed to non-revoked users whenever changes occur.

In the context of the architecture proposed in this work, a first strategy to be analyzed would be blockchain-based revocation state management. Since the blockchain layer is designed to store device records, attribute assignments, access policies, and key provisioning events, each attribute or key grant could be associated with lifecycle metadata, such as active, expired, revoked, or replaced. Before generating a new User Secret Key, the Attribute Authority would consult the smart contract to verify if the attributes required by the access policy are still valid. This strategy is compatible with the current on-chain/off-chain architecture, as the blockchain would remain responsible for auditability and state recording, while the Attribute Authority would remain responsible for cryptographic operations. However, this approach mainly prevents the issuance of new keys and does not immediately invalidate cryptographic material that has already been delivered to consumers.

A second strategy to be investigated would be the adoption of attributes with a limited lifespan or policy versioning. In this case, access policies would include a temporal or versioned component, such as an epoch identifier, and consumers would need periodically renewed keys to continue decrypting newly published data. For example, a policy could require not only traditional company attributes, such as function and department, but also a current version of the policy or a validity period. When an attribute is revoked, the policy version associated with the MQTT topic could be updated on the blockchain, and subsequent ciphertexts would be encryp-

ted under the new version. Revoked consumers would no longer receive updated keys from the Attribute Authority. This approach is related to version-based revocation strategies discussed by Peñuelas-Angulo et al. (2024), who review different revocation mechanisms for fog-enabled IoT, including partial key deletion, data owner revocation lists, version-based revocation, and time-based revocation.

A third strategy would involve more advanced mechanisms such as proxy re-encryption, third-party ciphertext updates, or revocable CP-ABE schemes with explicit support for attribute-level updates. Lu et al. (2024) propose a blockchain-based and CP-ABE-based access control scheme, in which the blockchain is used to track the distribution of private keys of attributes, while proxy re-encryption is used to update access rights. The proposed scheme addresses granular revocation by updating the symmetric key used to encrypt files, the attribute private keys used to protect access to symmetric keys, and the corresponding attribute ciphertext components. This type of approach is relevant to the proposed architecture because it suggests that the blockchain can be used not only as an audit layer for key provisioning events, but also as a mechanism to track key distribution and coordinate permission updates. However, adopting this strategy would add architectural and computational complexity, as it would require additional workflows for re-encryption, version synchronization, and experimental analysis of the overhead introduced by revocation events.

7.3.2 Scope of Cryptographic Scalability Assessment

The scalability assessment was restricted to experimental parameters directly associated with the cryptographic layer and the local message processing pipeline. In this sense, scalability was not treated as the architecture's ability to support a large number of simultaneous devices, organizations, or message streams. Instead, it was considered as the behavior of the cryptographic component when the size of the protected data, the number of attributes in the ABE policy, and the volume of individually protected messages increase.

The results indicate that, within the evaluated experimental scope, cryptographic processing time is more directly influenced by policy complexity than by payload size. However, the spatial overhead results show that payload size has a relevant impact on the relative increase in data volume, especially for small sensor measurements. This distinction is important because an architecture can exhibit stable processing times even while producing significant communication and storage overhead.

The assessment with real ESP32/DHT22 measurements showed that the accumulated overhead grows linearly when each measurement is individually protected. Therefore, the results provide

evidence about the behavior of the proposed solution under controlled message volume, but do not represent a complete assessment of system-level scalability.

Comprehensive system-level scalability analyses, involving distributed deployments, multiple organizations, thousands of simultaneous devices, or throughput under high-frequency industrial workloads, were not conducted at this stage and are outside the scope of this dissertation.

Chapter 8

CONCLUSIONS

Concluding this dissertation, this chapter gathers and summarizes the main findings obtained during the execution of this research. It will highlight the contributions to academia and industry, outlining a direction for potential future work and presenting the results produced during the development of the experiments.

8.1 Conclusions

The experimental results indicate that the proposed solution is technically feasible within the evaluated scope and capable of supporting fine-grained privacy control over shared data. The cryptographic evaluation showed that access policy complexity has a direct impact on processing time and spatial overhead, while payload size has a comparatively lower influence on processing time within the tested range.

The proposed model demonstrated the feasibility of allowing distinct organizations to share sensor data while controlling which attributes are required for data consumption. When combined with Hyperledger Besu, the architecture provides support for device registration, attribute management, policy traceability, and auditable access-related events. Overall, the tests performed demonstrated that the integration of MQTT, Attribute-Based Encryption, and blockchain-supported governance can support secure cross-organizational IIoT data sharing in a controlled experimental environment.

8.2 Contributions to Industry and Academia

From an academic perspective, this research contributes by proposing and testing a data sharing model for multi-enterprise scenarios in the IIoT. The work combines MQTT-based communication, attribute-based cryptography, and blockchain-supported governance to address

interoperability between independent organizations. Furthermore, the systematic literature review conducted during the research, in addition to providing an understanding of the event horizon of the topic, served as a foundation for identifying gaps related to granular access control, privacy preservation, interoperability, and architecture design for experimental evaluation of cryptographic overhead in IIoT environments.

Beyond the dissertation development, partial results of this research were peer-reviewed and published in an international scientific event. The article titled "Blockchain and Attribute-Based Encryption for Cross-Organizational IIoT" was published in the proceedings of the 16th IEEE International Conference on Industrial Applications (INDUSCON) 2025, DOI: 10.1109/INDUSCON66435.2025.11241288. This publication contributed to the scientific dissemination of the proposed approach and provided external peer validation of part of the research developed in this dissertation.

From an industrial perspective, this research contributes by exploring the use of established technologies, such as MQTT and Ethereum-compatible blockchain platforms, characteristics that increase the likelihood of adoption in real-world industrial environments. The proposed approach can support scenarios such as supply chain monitoring, collaborative manufacturing, and maintenance services, where organizations need to exchange operational data while preserving confidentiality, traceability, and control over access policies.

8.3 Future Work

One important direction for future work is evaluating the proposed solution in real-world industrial environments, where edge devices, gateways, and industrial controllers may present different computational capacities, memory constraints, and communication conditions.

Future work should also expand the current evaluation of cryptographic scalability. Large-scale deployment aspects, such as thousands of concurrent devices, multiple organizations, high-frequency message streams and distributed orchestration, were outside the experimental scope of this work and should be investigated only in future studies specifically designed for this purpose.

Another interesting direction involves attribute revocation. Based on the strategies discussed in Chapter 7, a future implementation of the architecture could contemplate recording the status of attributes and revocation events on the blockchain, thus allowing the Attribute Authority to verify the current state of each attribute before issuing new User Secret Keys. Additional mechanisms, such as attributes with expiration times, policy epochs, and proxy

re-encryption, could also be technically evaluated experimentally, as they may increase key management overhead, blockchain transactions, and cryptographic processing costs.

Another direction for future work involves investigating the integration of the proposed ABE-based cryptographic workflow with broader scenarios of blockchain interoperability and post-quantum security. In the current prototype, Hyperledger Besu is used to record policies, key-related metadata, and audit events, while cryptographic operations remain off-chain, and from this perspective, future studies could assess how this model behaves with post-quantum ABE schemes. Recent work has explored data exchange between post-quantum blockchains based on ABE using multi-layered blockchain architectures and IPFS (YU; MU, 2024). While these approaches are outside the scope of the current prototype, they point to possible research directions for extending the proposed framework.

8.4 Research Artifact

As an outcome of this research, a prototype of the proposed architecture was implemented and experimentally validated. The artifact was designed to support reproducibility and to document the main implementation components used in the evaluation of the proposed solution.

The research artifact is available as a public repository containing the implementation of the proof of concept, experimental scripts, smart contracts, reproducibility documentation, and supporting evidence files. The artifact includes the integrated proof-of-concept module, composed of the `<AccessPolicyRegistry>` smart contract, Python orchestration scripts, OpenABE integration components, ECIES-based key protection utilities, subscriber-side key recovery logic, and a Streamlit-based interface for guided validation.

The project comprises:

- An Attribute-Based Encryption (ABE) cryptographic module implemented using the OpenABE library.
- The `<AccessPolicyRegistry>` smart contract, responsible for registering devices, topic policies, subscriber attributes, access requests, encrypted key grants, and integrity hashes.
- Python orchestration scripts for on-chain bootstrap, MQTT sensor data collection, ABE encryption, access request registration, Attribute Authority processing, subscriber-side key retrieval, and payload decryption.

- ECIES-based utilities for protecting the delivery of OpenABE User Secret Keys (USKs) to subscribers.
- Performance evaluation scripts for encryption and decryption operations, enabling controlled experimental analysis of processing time and spatial overhead.
- Deployment and validation procedures for a private Hyperledger Besu blockchain network used to manage authorization workflows and access-related metadata.
- A Streamlit-based interface for guided execution, inspection of evidence files, and validation of authorized and unauthorized access scenarios.
- Technical documentation containing installation guidelines, system architecture description, dependency configuration, and reproducibility instructions.

The repository also documents the reproduction workflow for authorized and unauthorized access scenarios, including environment verification, on-chain bootstrap, MQTT sensor data collection, ABE encryption, access request registration, Attribute Authority processing, encrypted key retrieval, integrity verification, and payload decryption. This documentation supports the verification of the experimental flow and enables future extensions of the artifact.

The complete source code and developer documentation are maintained in the following public repository and documentation page:

Technical documentation: <https://fiorezewesley.github.io/cross-org-iiot-security-blockchain/>

Source code repository: <https://github.com/fiorezewesley/cross-org-iiot-security-blockchain>

REFERENCES

- ABUSERRIEH, L.; ALALFI, M. H. A survey on verification of security and safety in iot systems. *IEEE Access*, IEEE, v. 12, p. 1–16, 2024. Available at: [⟨https://ieeexplore.ieee.org/document/3413071⟩](https://ieeexplore.ieee.org/document/3413071).
- AFRIN, S.; RAFA, S. J.; KABIR, M.; FARAH, T.; ALAM, M. S. B.; LAMEESA, A.; AHMED, S. F.; GANDOMI, A. H. Industrial internet of things: Implementations, challenges, and potential solutions across various industries. *Computers in Industry*, v. 170, p. 104317, 2025. ISSN 0166-3615. Available at: [⟨https://www.sciencedirect.com/science/article/pii/S016636152500082X⟩](https://www.sciencedirect.com/science/article/pii/S016636152500082X).
- ALHARBY, M.; MOORSEL, A. van. Blockchain-based smart contracts: A systematic mapping study. *Computer Science & Information Technology*, AIR Publishing, p. 125–140, 2017.
- ALSOLAMI, R. A.; CHAUHDARY, S. H.; ALSHDADI, A. A. Interoperable access framework for internet of things backhauled heterogeneous applications. *Microprocessors and Microsystems*, Elsevier B.V., v. 82, April 2021.
- BAYRAMCAVUS, A.; KAYA, M. Çağrı; DOĞRU, A. H. Interoperability of microservice-based systems. In: *Proceedings of the 13th International Conference on Electrical and Electronics Engineering (ELECO)*. Virtual, Bursa, Türkiye: IEEE, 2021. Available at: [⟨https://www.scopus.com/inward/record.uri?partnerID=HzOxMe3b&scp=85125222570&origin=inward⟩](https://www.scopus.com/inward/record.uri?partnerID=HzOxMe3b&scp=85125222570&origin=inward).
- CACHIN, C.; VUKOLIC, M. Blockchain consensus protocols in the wild. *arXiv preprint arXiv:1707.01873*, 2017. Available at: [⟨https://arxiv.org/abs/1707.01873⟩](https://arxiv.org/abs/1707.01873).
- CAPOCASALE, V. et al. Comparative analysis of permissioned blockchains: Performance and privacy features with Hyperledger Besu. *Journal of Information Security and Applications*, v. 79, p. 102992, 2023. Available at: [⟨https://www.sciencedirect.com/science/article/pii/S2096720922000549⟩](https://www.sciencedirect.com/science/article/pii/S2096720922000549).
- Centre for Reviews and Dissemination. *Centre for Reviews and Dissemination - University of York*. 2024. Accessed: 2024-07-07. Available at: [⟨https://www.york.ac.uk/media/crd/Systematic_Reviews.pdf⟩](https://www.york.ac.uk/media/crd/Systematic_Reviews.pdf).
- Check Point Research. *The Tipping Point: Exploring the Surge in IoT Cyberattacks Globally*. 2024. Accessed: 2024-07-04. Available at: [⟨https://blog.checkpoint.com/security/the-tipping-point-exploring-the-surge-in-iotcyberattacks-plaguing-the-education-sector/⟩](https://blog.checkpoint.com/security/the-tipping-point-exploring-the-surge-in-iotcyberattacks-plaguing-the-education-sector/).
- CHEN, T.; YU, Y.; DUAN, Z.; GAO, J.; LAN, K. Blockchain/abe-based fusion solution for e-government data sharing and privacy protection. In: *ASSOCIATION FOR COMPUTING MACHINERY. ACM International Conference Proceeding Series*. [S.l.], 2020.

- CUI, H.; CHEN, Z.; YU, X.; CHEN, H.; HAO, J. Iot data management and lineage traceability: A blockchain-based solution. In: IEEE INC. 2019 IEEE/CIC International Conference on Communications Workshops in China (ICCC Workshops). Beijing, China: IEEE, 2019. p. 239–244.
- DASAKLIS, T. K.; CASINO, F.; PATSAKIS, C. Defining granularity levels for supply chain traceability based on iot and blockchain. In: *Proceedings of the International Conference on Omni-Layer Intelligent Systems (COINS)*. Crete, Greece: Association for Computing Machinery, 2019. p. 1–7. ISBN 978-1-4503-6640-3.
- DING, N.; ZHANG, Y.; YU, W.; WANG, Y.; GUO, Z. Economic incentive mechanisms for blockchain-based iot: A comprehensive survey. *IEEE Internet of Things Journal*, IEEE, v. 8, n. 7, p. 5743–5763, 2021. Available at: <https://par.nsf.gov/servlets/purl/10280174>.
- DOMINGUES, M.; FARIA, J. N.; PORTUGAL, D. Dimensioning payload size for fast retransmission of mqtt packets in the wake of network disconnections. *EURASIP Journal on Wireless Communications and Networking*, 2024.
- ENGSTRÖM, E.; STOREY, M.-A.; RUNESON, P.; HÖST, M.; BALDASSARRE, M. T. How software engineering research aligns with design science: a review. *Empirical Software Engineering*, Springer, v. 25, p. 2630–2660, 2020. Available at: <https://doi.org/10.1007/s10664-020-09818-7>.
- European Commission. *New European Interoperability Framework: Promoting Seamless Services and Data Flows for European Public Administrations*. Luxembourg, 2017. Available at: <https://doi.org/10.2799/78681>.
- EVANS, P. C.; ANNUNZIATA, M. *Industrial Internet: Pushing the Boundaries of Minds and Machines*. [S.l.], 2012. Published on November 26, 2012.
- FAN, S.; LIN, C.; KHAZAEI, H.; MUSILEK, P. Performance analysis of hyperledger Besu in private blockchain. In: 2022 IEEE International Conference on Decentralized Applications and Infrastructures (DAPPS). [s.n.], 2022. p. 64–73. Available at: <https://doi.org/10.1109/DAPPS.2022.9899854>.
- FERNANDEZ-GAGO, C.; FERRARIS, D.; ROMAN, R.; LOPEZ, J. Trust interoperability in the internet of things. *Internet of Things*, v. 26, p. 101226, 2024. ISSN 2542-6605. Available at: <https://www.sciencedirect.com/science/article/pii/S2542660524001677>.
- GAVRIILIDIS, N. O.; HALKIDIS, S. T.; PETRIDOU, S. Empirical evaluation of tls-enhanced mqtt on iot devices for v2x use cases. *Applied Sciences*, v. 15, n. 15, 2025. ISSN 2076-3417. Available at: <https://www.mdpi.com/2076-3417/15/15/8398>.
- GREGOR, S.; HEVNER, A. R. The front end of innovation: Perspectives on creativity, knowledge and design. *Lecture Notes in Computer Science*, v. 9073, p. 249–263, 2015.
- GUGUEOTH, V.; SAFAVAT, S.; SHETTY, S.; RAWAT, D. A review of IoT security and privacy using decentralized blockchain techniques. *Computer Science Review*, Elsevier Ireland Ltd, November 2023.
- GUO, R. et al. Flexible and efficient blockchain-based ABE scheme with multi-authority for medical on demand in telemedicine system. *IEEE Access*, v. 7, p. 88012–88025, 2019.

- HAMAD, M.; FINKENZELLER, A.; LIU, H.; LAUNGER, J.; PREVELAKIS, V.; STEINHORST, S. Seemqtt: Secure end-to-end mqtt-based communication for mobile iot systems using secret sharing and trust delegation. *IEEE Internet of Things Journal*, v. 10, n. 4, p. 3384–3406, 2023. Available at: <https://doi.org/10.1109/JIOT.2022.3221857>.
- HAZRA, A.; ADHIKARI, M.; AMGOTH, T.; SRIRAMA, S. N. A comprehensive survey on interoperability for iiot: Taxonomy, standards, and future directions. *ACM Computing Surveys (CSUR)*, Association for Computing Machinery, v. 55, n. 1, p. 1–35, 2021.
- HELU, M.; SPROCK, T.; HARTENSTINE, D.; VENKETESH, R.; SOBEL, W. Scalable data pipeline architecture to support the industrial internet of things. *CIRP Annals - Manufacturing Technology*, Elsevier, v. 69, p. 385–388, 2020.
- HEVNER, A.; R, A.; MARCH, S.; T, S.; PARK; PARK, J.; RAM; SUDHA. Design science in information systems research. *Management Information Systems Quarterly*, v. 28, p. 75–, 03 2004.
- HUR, J.; NOH, D. K. Attribute-based access control with efficient revocation in data outsourcing systems. *IEEE Transactions on Parallel and Distributed Systems*, v. 22, n. 7, p. 1214–1221, 2011. Available at: <https://doi.org/10.1109/TPDS.2010.203>.
- HUSSEIN, M.; SARHAN, S.; ABUELENIN, S. M. Consensus algorithms in blockchain: Comparative analysis and performance evaluation. *Cybersecurity*, SpringerOpen, v. 6, n. 1, p. 1–24, 2023. Available at: <https://cybersecurity.springeropen.com/articles/10.1186/s42400-023-00163-y>.
- Industrial Internet Consortium. *Industrial Internet of Things Volume G1: Reference Architecture*. Needham, MA, USA: Industrial Internet Consortium, 2017.
- INTERNATIONAL ORGANIZATION FOR STANDARDIZATION. *ISO 11354-1:2011 – Advanced Automation Technologies and Their Applications – Requirements for Establishing Manufacturing Enterprise Process Interoperability – Part 1: Framework for Enterprise Interoperability*. Geneva, Switzerland, 2011.
- KAGERMANN, H.; WAHLSTER, W.; HELBIG, J. *Recommendations for Implementing the Strategic Initiative INDUSTRIE 4.0*. Munich, Germany: acatech – National Academy of Science and Engineering, 2013.
- KARANKAR, N.; SETH, A. An iot system for access control using blockchain and message queuing system. *EURASIP Journal on Information Security*, v. 2025, n. 1, p. 31, 2025. ISSN 2510-523X. Available at: <https://doi.org/10.1186/s13635-025-00208-4>.
- KHAN, A. A.; BOUROUIS, S.; KAMRUZZAMAN, M. M.; HADJOUNI, M.; SHAIKH, Z. A.; LAGHARI, A. A.; ELMANNAI, H.; DHAHBI, S. Data security in healthcare industrial internet of things with blockchain. *IEEE Sensors Journal*, v. 23, p. 25144–25151, 2023. Available at: <https://api.semanticscholar.org/CorpusID:258648530>.
- KHAN, S. N.; LOUKIL, F.; GHEDIRA-GUEGAN, C.; BENKHELIFA, E.; BANI-HANI, A. Blockchain smart contracts: Applications, challenges, and future trends. *Peer-to-Peer Networking and Applications*, v. 14, n. 5, p. 2901–2925, 2021.

- KIM, S.-K.; VONG, H. C. Secured network architectures based on blockchain technologies: A systematic review. *ACM Computing Surveys*, Association for Computing Machinery, New York, NY, USA, v. 57, n. 7, 2025. ISSN 0360-0300. Available at: [⟨https://doi.org/10.1145/3715000⟩](https://doi.org/10.1145/3715000).
- KITCHENHAM, B.; BRERETON, P. A systematic review of systematic review process research in software engineering. *Information and Software Technology*, Elsevier B.V., 2013. Available at: [⟨https://www.sciencedirect.com/science/article/pii/S0950584913001560⟩](https://www.sciencedirect.com/science/article/pii/S0950584913001560).
- LIN, I.-C.; LIAO, T.-C. A survey of blockchain security issues and challenges. *International Journal of Network Security*, v. 19, n. 5, p. 653–659, 2017.
- LOSS, S.; CACHO, N.; LOPES, F. Blockchain strategy for multi-level interoperability in public safety scenario. In: ASSOCIATION FOR COMPUTING MACHINERY. *ACM International Conference Proceeding Series*. [S.l.], 2023.
- LU, Y.; FENG, T.; LIU, C.; ZHANG, W. A blockchain and cp-abe based access control scheme with fine-grained revocation of attributes in cloud health. *Computers, Materials & Continua*, v. 78, n. 2, p. 2787–2810, 2024. Available at: [⟨https://doi.org/10.32604/cmc.2023.046106⟩](https://doi.org/10.32604/cmc.2023.046106).
- MAHALLE, P. N.; BHATTACHARYA, S.; RAKESH, N.; TATIYA, M. R.; GULHANE, M.; SHERJE, N. Building a unified framework to standardize and improve interoperability in iot systems. In: *Proceedings of the 6th International Conference on Information Management & Machine Intelligence*. New York, NY, USA: Association for Computing Machinery, 2025. (ICIMMI '24). ISBN 9798400711220. Available at: [⟨https://doi.org/10.1145/3745812.3745887⟩](https://doi.org/10.1145/3745812.3745887).
- MAHENDRAN, R. K.; KHAN, A.; ULLAH, F.; ALI, F.; ALZUBI, A. A. Prism-iiot: A holistic approach for privacy preservation in industrial iot using advanced cryptography and blockchain-enabled auditability framework. *Alexandria Engineering Journal*, v. 128, p. 816–832, 2025. ISSN 1110-0168. Available at: [⟨https://www.sciencedirect.com/science/article/pii/S1110016825008506⟩](https://www.sciencedirect.com/science/article/pii/S1110016825008506).
- MISHRA, R. K.; YADAV, R. K.; NATH, P. Blockchain powered iot access control model for secure data sharing and management: Performance analysis. In: INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS INC. *Proceedings of the 2023 2nd International Conference on Informatics (ICI 2023)*. [S.l.], 2023.
- NISHI, H.; WEJEKOON, J.; SONG, E. Y.; LEE, K. B. Security for ieeep1451.1.6-based sensor networks for iot applications. In: *Proceedings of the 50th Annual Conference of the IEEE Industrial Electronics Society (IECON 2024)*. [S.l.]: IEEE, 2024.
- OUADDAH, A.; ELKALAM, A. A.; OUAHMAN, A. A. Towards a novel privacy-preserving access control model based on blockchain technology, attribute-based encryption, and self-sovereign identity. *Computer Networks*, Elsevier, v. 186, p. 107735, 2021. Available at: [⟨https://doi.org/10.1016/j.comnet.2020.107735⟩](https://doi.org/10.1016/j.comnet.2020.107735).
- PAUL, J.; LIM, W. M.; O'CASS, A.; HAO, A.; BRESCIANI, S. Scientific procedures and rationales for systematic literature reviews (spar-4-slr). *International Journal of Consumer Studies*, Wiley, 2021.

- PEFFERS, K.; TUUNANEN, T.; GENGLER, C. E.; ROSSI, M.; HUI, W.; VIRTANEN, V.; BRAGGE, J. The design science research process: A model for producing and presenting information systems research. In: *Proceedings of the First International Conference on Design Science Research in Information Systems and Technology (DESRIST)*. [S.l.: s.n.], 2006.
- PEFFERS, K.; TUUNANEN, T.; ROTHENBERGER, M.; CHATTERJEE, S. A design science research methodology for information systems research. *Journal of Management Information Systems*, v. 24, p. 45–77, 01 2007.
- PEÑUELAS-ANGULO, A.; FERREGRINO-URIBE, C.; MORALES-SANDOVAL, M. A revocable multi-authority attribute-based encryption scheme for fog-enabled iot. *Journal of Systems Architecture*, v. 155, p. 103265, 2024. Available at: <https://doi.org/10.1016/j.sysarc.2024.103265>.
- PERAZZO, P.; RIGHETTI, F.; MANNA, M. L.; VALLATI, C. Performance evaluation of attribute-based encryption on constrained iot devices. *Computer Communications*, v. 170, p. 151–163, 2021. Available at: <https://doi.org/10.1016/j.comcom.2021.02.012>.
- PETER, O.; PRADHAN, A.; MBOHWA, C. Industrial internet of things (IIoT): Opportunities, challenges, and requirements in manufacturing businesses in emerging economies. In: ELSEVIER B.V. *Procedia Computer Science*. [S.l.], 2022.
- QIU, J.; TIAN, Z.; DU, C.; ZUO, Q.; SU, S.; FANG, B. A survey on access control in the age of internet of things. *IEEE Internet of Things Journal*, v. 7, n. 6, p. 4682–4696, 2020. Available at: <https://doi.org/10.1109/JIOT.2020.2969326>.
- RASORI, M.; MANNA, M. L.; PERAZZO, P.; DINI, G. A survey on attribute-based encryption schemes suitable for the internet of things. *IEEE Internet of Things Journal*, v. 9, n. 11, p. 8269–8290, 2022. Available at: <https://doi.org/10.1109/JIOT.2022.3154039>.
- RATH, C. K.; MANDAL, A. K.; SARKAR, A. Microservice based scalable iot architecture for device interoperability. *Computer Standards Interfaces*, v. 84, p. 103697, 2023. ISSN 0920-5489. Available at: <https://www.sciencedirect.com/science/article/pii/S0920548922000642>.
- REN, Z.; ZHANG, L.; CHEN, X.; ZHAO, Y.; LIU, J.; YANG, J. Blockchain-based cp-abe data sharing and privacy-preserving scheme using distributed kms and zero-knowledge proof. *Journal of King Saud University - Computer and Information Sciences*, Elsevier, v. 36, n. 2, p. 155–168, 2024. Available at: <https://www.sciencedirect.com/science/article/pii/S1319157824000582>.
- SALEEM, J.; RAZA, U.; HOLDERBAUM, W. Transforming industry 4.0 security: Analysis of abe and aba technologies. *IECE Transactions on Intelligent Systematics*, Institute of Electronics, Information and Communication Engineers, v. 1, n. 3, p. 127–144, 2024.
- SHAKARAMI, M.; BENSON, J.; SANDHU, R. Blockchain-based administration of access in smart home iot. In: *Proceedings of the 2022 ACM Workshop on Secure and Trustworthy Cyber-Physical Systems (SaT-CPS '22)*. Baltimore, MD, USA: Association for Computing Machinery, 2022. p. 57–66. ISBN 978-1-4503-9229-7.
- SINGAMANENI, K. K.; YADAV, K.; ALEDAILY, A.; VIRIYASITAVAT, W.; DHIMAN, G.; KAUR, A. Decoding the future: exploring and comparing abe standards for cloud, iot,

- blockchain security applications. *Multimedia Tools and Applications*, v. 84, p. 12299–12327, 05 2024.
- SOLTANI, R.; NGUYEN, U. T.; AN, A. A survey of self-sovereign identity ecosystem. *Security and Communication Networks*, Wiley, v. 2021, p. 1–31, 2021. Available at: <https://doi.org/10.1155/2021/8873429>.
- SYCHOWIEC, J.; ZIELINSKI, Z. An experimental framework for secure and reliable data streams distribution in federated iot environments. In: INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS INC. *Proceedings of the 18th Conference on Computer Science and Intelligence Systems (FedCSIS 2023)*. [S.l.], 2023.
- TRIPATHI, G.; AHAD, M.; CASALINO, G. A comprehensive review of blockchain technology: Underlying principles and historical background with future challenges. *Journal of Network and Computer Applications*, Elsevier, v. 209, p. 104494, 2023. Available at: <https://www.sciencedirect.com/science/article/pii/S2772662223001844>.
- TSCHORSCH, F.; SCHEUERMANN, B. Bitcoin and beyond: A technical survey on decentralized digital currencies. *IEEE Communications Surveys Tutorials*, v. 18, n. 3, p. 2084–2123, 2016.
- UNTERKALMSTEINER, M.; GORSCHKE, T.; ISLAM, A. K. M. M.; CHENG, C. K.; PERMADI, R. B.; FELDT, R. Evaluation and measurement of software process improvement—a systematic literature review. *IEEE Transactions on Software Engineering*, v. 38, n. 2, p. 398–424, mar. 2012. Available at: <https://doi.org/10.1109/TSE.2011.26>.
- W3C. *Verifiable Credentials Data Model 2.0*. 2025. Accessed: 2025-07-04. Available at: <https://www.w3.org/TR/vc-data-model-2.0/>.
- WANG, X.; ZHANG, J.; SCHOOLER, E. M.; ION, M. Performance evaluation of attribute-based encryption: Toward data privacy in the iot. In: *2014 IEEE International Conference on Communications (ICC)*. [s.n.], 2014. p. 725–730. Available at: <https://doi.org/10.1109/ICC.2014.6883405>.
- WU, C.-Y.; HUANG, K.-H.; HSU, C.-Y. A decentralised multi-authority attribute-based encryption for secure and scalable iot access control. *Applied Sciences*, MDPI, v. 15, n. 7, p. 3890, 2025. Available at: <https://doi.org/10.3390/app15073890>.
- XU, R.; CHEN, Y.; BLASCH, E.; CHEN, G. A federated capability-based access control mechanism for internet of things (iots). In: *Proceedings of SPIE*. [s.n.], 2018. Available at: <https://doi.org/10.1117/12.2305619>.
- YU, H. F.; MU, W. Z. ABE-based postquantum cross-blockchain data exchange approach for smart agriculture. *IEEE Transactions on Industrial Informatics*, 2024.
- Zeutro LLC. *OpenABE: Attribute-Based Encryption Library*. [S.l.], 2019. Available at: <https://github.com/zeutro/openabe>.
- ZEYDAN, E.; BICEN, A.; ERPEK, T.; ALAGOZ, F. Blockchain-based peer-to-peer networks: Fundamentals and future research directions. In: *Proceedings of the 2023 5th International Conference on Blockchain Technology (ICBCT)*. New York, NY, USA: Association for Computing Machinery, 2023. p. 46–51. Available at: <https://dl.acm.org/doi/10.1145/3756330>.

ZHANG, J.; XIN, Y.; GAO, Y.; LEI, X.; YANG, Y. Secure abe scheme for access management in blockchain-based iot. *IEEE Access*, Institute of Electrical and Electronics Engineers, v. 9, p. 54840–54849, 2021. Available at: <https://doi.org/10.1109/ACCESS.2021.3071031>.

ZIEGLER, D.; SABONGUI, J.; PALFINGER, G. Fine-grained access control in industrial internet of things: Evaluating outsourced attribute-based encryption. In: *Proceedings of the IFIP International Information Security and Privacy Conference (IFIP SEC)*. [S.l.]: Springer, 2019. p. 91–104.